Dorey's Resource Book

Topic 1 – Differential Calculus

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1 The derivative of $e^{f(x)}$

ANCHOR QUESTION

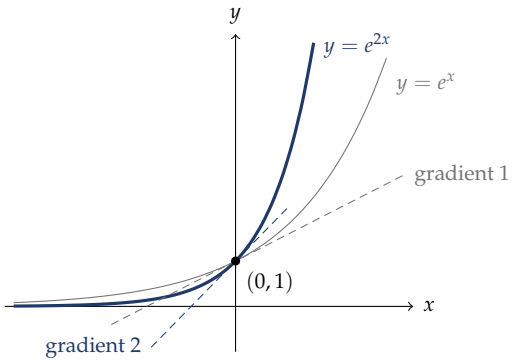
Differentiate $y = e^{x^2-5x}$.

SYLLABUS MAV-12-04

- Apply the rules of differentiation to find the derivative of $f(x) = ke^{ax}$, where k and a are constants
- Use the chain rule to prove $\frac{d}{dx}\left(e^{ax+b}\right) = ae^{ax+b}$, where a and b are constants and $a \neq 0$
- Use the chain rule to differentiate functions of the form $e^{f(x)}$

NOTES

- $\frac{d}{dx}\left(e^x\right) = e^x$.
- $\frac{d}{dx}\left(e^{f(x)}\right) = f'(x)\,e^{f(x)}$. This is on the reference sheet.
- The power stays exactly the same. Multiply by the derivative of the power.
- Special case: $\frac{d}{dx}\left(e^{ax+b}\right) = a\,e^{ax+b}$, and $\frac{d}{dx}\left(ke^{ax}\right) = ka\,e^{ax}$.
- Proof: let $u = ax + b$, so $y = e^u$. Then $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = e^u \times a = ae^{ax+b}$.
- x in the power of e: use this rule. x in the base, as in x^e: use the power rule. No x, as in e^4: a constant, derivative 0.
- Rewrite first with index laws: $\frac{1}{e^{kx}} = e^{-kx}$, $e^a e^b = e^{a+b}$, $\sqrt{e^x} = e^{x/2}$.
- $e^{f(x)} > 0$ for every x, so $f'(x)\,e^{f(x)} = 0$ only when $f'(x) = 0$.



WORKED EXAMPLE

Differentiate $y = e^{3x^2-x}$.

$f(x) = 3x^2 - x$, so $f'(x) = 6x - 1$. the power and its derivative

$$\frac{dy}{dx} = f'(x)\,e^{f(x)}$$
$$= (6x - 1)\,e^{3x^2-x}$$

Tier 1 · the derivative of $e^{f(x)}$

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1. For each function, write the power $f(x)$, then find $f'(x)$.

a) e^{9x}

b) e^{2-x}

c) e^{x^4-3x}

2. Differentiate:

a) $y = e^{7x}$

b) $y = e^{x^2+1}$

c) $y = e^{x^3-6x}$

3. Differentiate:

a) $y = e^{1-5x}$

b) $y = e^{\sqrt{x}}$

c) $y = e^4$

Tier 2 · a coefficient in front

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4. Differentiate:

a) $y = 4e^{3x}$

b) $y = -2e^{5x+1}$

c) $y = \frac{1}{2}e^{6x}$

5. Differentiate:

a) $y = 3e^{x^2}$

b) $y = -e^{2x^2-x}$

6. Show that $\dfrac{d}{dx}\left(\frac{1}{4}e^{8x-2}\right) = 2e^{8x-2}$.

Tier 3 · sums and differences

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7. Differentiate:

a) $y = e^{4x} + e^{-4x}$

b) $y = 3e^{2x} - x^3$

8. Differentiate $y = 2e^{-3x} - 5e^{x} + 7x$.

9. Differentiate $y = x^2 - 4x + e^{2-x}$.

Tier 4 · rewrite with index laws first

10. Differentiate:

a) $y = \dfrac{1}{e^x}$

b) $y = \dfrac{4}{e^{2x}}$

11. Differentiate:

a) $y = e^x \cdot e^{3x}$

b) $y = \left(e^{2x}\right)^3$

12. Differentiate:

a) $y = \sqrt{e^{4x}}$

b) $y = (e^x + e^{-x})^2$, by expanding first

Tier 5 · exact values of the derivative

13. If $f(x) = e^{2x+1}$, find the exact value of $f'(0)$.

14. If $f(x) = e^{x^2-4}$, find the exact value of $f'(2)$.

15. Find the gradient of the tangent to $y = 3e^{-x} + e^x$ at the point where $x = \ln 3$.

Tier 6 · the chain rule on the outside

16. Differentiate $y = (e^x + 1)^3$.

17. Differentiate $y = \left(2 - e^{3x}\right)^4$.

18. Differentiate $y = \sqrt{1 + e^{2x}}$.

Tier 7 · show that, then hence

19. Let $y = e^{x^2} \cdot e^{-2x}$.

a) Show that $\dfrac{dy}{dx} = 2(x-1)\,e^{x^2-2x}$.

b) Hence find the exact coordinates of the point where the tangent is horizontal.

20. Let $y = \dfrac{e^{2x^2}}{e^{8x}}$.

a) Show that $\dfrac{dy}{dx} = 4(x-2)\,e^{2x^2-8x}$.

b) Hence find the exact coordinates of the point where the tangent is horizontal.

21. Let $f(x) = e^{x^3-3x}$.

a) Find $f'(x)$ in fully factorised form.

b) Find the exact coordinates of the points where the tangent is horizontal.

Conceptual

22. A student writes $\frac{d}{dx}\left(e^{3x}\right) = 3x\,e^{3x-1}$. Explain the error and give the correct derivative.

23. Use the chain rule to prove that $\frac{d}{dx}\left(e^{ax+b}\right) = ae^{ax+b}$, where a and b are constants.

24. Explain why the graph of $y = 5e^{2x}$ is increasing for every value of x.

25. The curve $y = e^{x^2+kx}$ has a horizontal tangent at $x = 3$. Find the value of k.

26. Let $y = 5e^{2x}$.

(a) Find $\frac{dy}{dx}$. (1 mark)

(b) Show that $\frac{dy}{dx} = 2y$. (1 mark)

(c) Hence find the gradient of the curve at the point where $y = 10$. (1 mark)

Revision

27. Let $f(x) = x(2x-1)^4$.

(a) Show that $f'(x) = (2x-1)^3(10x-1)$. (2 marks)

(b) Find the equation of the tangent to $y = f(x)$ at the point where $x = 1$. (2 marks)

(c) Find the values of x where the tangent to $y = f(x)$ is horizontal. (1 mark)

28.

(a) Solve $\ln x - \frac{2}{\ln x} = 1$. (3 marks)

(b) Given that $\log_a 2 = 0.36$ and $\log_a 3 = 0.57$, find $\log_a 12$. (1 mark)

(c) Using the same values, find $\log_a 1.5$. (1 mark)

29. Xena and Gabrielle compete in a series of games. The series finishes when one player has won two games. In any game, the probability that Xena wins is $\frac{3}{5}$ and the probability that Gabrielle wins is $\frac{2}{5}$.

(a) Draw a tree diagram showing all the possible outcomes of the series. (1 mark)

(b) What is the probability that Gabrielle wins the series? (2 marks)

(c) What is the probability that three games are played? (2 marks)

2 Derivatives of a^x, products and quotients

ANCHOR QUESTION

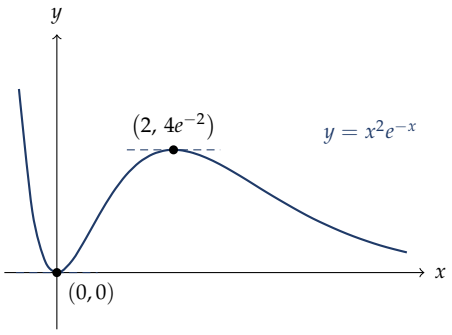
Differentiate $y = x^2e^{3x}$, giving your answer in factorised form.

SYLLABUS MAV-12-04

- Prove and use the formula $\frac{d}{dx}(a^x) = (\ln a)\,a^x$, where a is a constant and $a > 0$
- Apply the product, quotient and chain rules to differentiate functions of the form $f(x)g(x)$, $\frac{f(x)}{g(x)}$ or $f(g(x))$, where $f(x)$ and $g(x)$ are any of the functions within the scope of the Mathematics Advanced course

NOTES

- Any positive number is a power of e: $a = e^{\ln a}$, so $a^x = e^{x\ln a}$.
- $\frac{d}{dx}(a^x) = (\ln a)\,a^x$. Proof: $\frac{d}{dx}\left(e^{x\ln a}\right) = \ln a \cdot e^{x\ln a} = (\ln a)\,a^x$.
- $\ln a$ is just a number. When $a = e$, $\ln e = 1$ and the rule gives back $\frac{d}{dx}(e^x) = e^x$.
- Product rule: if $y = uv$, then $\frac{dy}{dx} = u'v + uv'$.
- Quotient rule: if $y = \frac{u}{v}$, then $\frac{dy}{dx} = \frac{u'v - uv'}{v^2}$. Both rules are on the reference sheet.
- After the product rule, both terms share the exponential: take it out as a common factor, e.g. $2xe^{3x} + 3x^2e^{3x} = xe^{3x}(2 + 3x)$.
- An exponential is never 0, so a factorised derivative is 0 only where its other factors are 0.



WORKED EXAMPLE

Differentiate $y = x^2e^{5x}$, giving your answer in factorised form.

$u = x^2,\ u' = 2x; \quad v = e^{5x},\ v' = 5e^{5x}$ — name the parts

$\frac{dy}{dx} = u'v + uv' = 2xe^{5x} + 5x^2e^{5x}$ — product rule

$= xe^{5x}(2 + 5x)$ — common factor xe^{5x}

Tier 1 · the product rule with e

30. Write as a power of e.

a) 9^x

b) 2^{5x}

31. Factorise fully.

a) $3x^2e^{2x} + 2x^3e^{2x}$

b) $e^{3x} - 3xe^{3x}$

32. Differentiate:

a) $y = 6^x$

b) $y = 5 \cdot 2^x$

c) $y = 3^{4x}$

33. Differentiate, giving each answer in factorised form.

a) $y = xe^x$

b) $y = x^2e^{4x}$

34. Differentiate $y = x^4e^x$, giving your answer in factorised form.

Tier 2 · a negative power

35. Differentiate $y = (3x - 1)e^{-x}$, giving your answer in factorised form.

36. Differentiate $y = xe^{-2x}$, giving your answer in factorised form.

37. Differentiate $y = x^2e^{-3x}$, giving your answer in factorised form.

Tier 3 · the quotient rule

38. Differentiate $y = \dfrac{e^{3x}}{x}$.

39. Differentiate $y = \dfrac{x}{e^{2x}}$, simplifying your answer.

40. Differentiate $y = \dfrac{e^x}{x+1}$.

Tier 4 · products with a^x

41. Differentiate $y = x \cdot 2^x$.

42. Differentiate $y = x^2 \cdot 5^x$, giving your answer in factorised form.

43. Differentiate $y = 2^x e^x$.

Tier 5 · quotients of exponentials

44. Differentiate $y = \dfrac{e^x}{e^x + 2}$.

45. Differentiate $y = \dfrac{e^x - 1}{e^x + 1}$.

46. Differentiate $y = \dfrac{e^{2x}}{1 + e^{2x}}$.

Tier 6 · horizontal tangents

47. Find the exact coordinates of the point on $y = xe^{-x}$ where the tangent is horizontal.

48. Find the exact coordinates of the points on $y = x^2 e^x$ where the tangent is horizontal.

49. Find the exact coordinates of the point on $y = \dfrac{e^x}{x}$ where the tangent is horizontal.

Conceptual

50. Prove that $\dfrac{d}{dx}\left(a^x\right) = (\ln a)\, a^x$, where $a > 0$.

51. A student writes $\dfrac{d}{dx}\left(2^x\right) = x \cdot 2^{x-1}$. Explain the error and give the correct derivative.

52. Show that $\dfrac{d}{dx}\left(xe^x\right) = (x+1)e^x$, and hence find the gradient of $y = xe^x$ at the origin.

53. Find the exact gradient of the tangent to $y = xe^{2x}$ at the point where $x = 1$.

54. Let $f(x) = xe^{-x}$.

(a) Show that $f'(x) = (1 - x)e^{-x}$. (1 mark)

(b) Explain why $f(x)$ is increasing for $x < 1$. (1 mark)

(c) Find the exact coordinates of the point where the tangent to $y = f(x)$ is horizontal. (1 mark)

Revision

55. Let $f(x) = e^{x^2-2x}$.

(a) Find $f'(x)$. (1 mark)

(b) Find the exact coordinates of the point where the tangent to $y = f(x)$ is horizontal. (2 marks)

(c) Find the equation of the tangent to $y = f(x)$ at the point where $x = 2$. (2 marks)

56. Consider the quadratic equation $x^2 - (k+2)x + (k+5) = 0$, where k is a constant.

(a) Find the values of k for which the equation has equal roots. (3 marks)

(b) Solve the equation when $k = 4$. (1 mark)

(c) For what values of k does the equation have no real roots? (1 mark)

57. The points A and B lie on a circle with centre O and radius 2 cm, and $\angle AOB = \theta$.

(a) There are two values of θ for which the area of triangle AOB is $\sqrt{3}$ cm^2. One is $\frac{\pi}{3}$. Find the other. (2 marks)

(b) Suppose $\theta = \frac{\pi}{3}$. Find the area of the sector AOB. (1 mark)

(c) Find the exact perimeter of the minor segment bounded by the chord AB and the arc AB. (2 marks)

3 Tangents and normals with exponentials

ANCHOR QUESTION

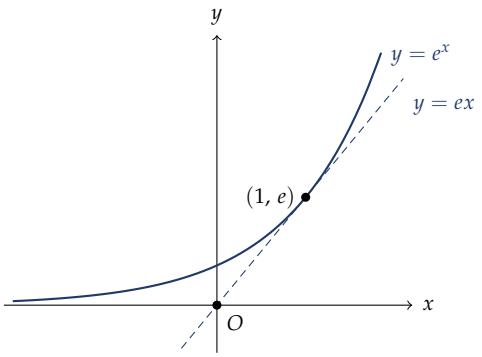
Find the equation of the tangent to $y = e^{2x}$ at the point where $x = 1$.

SYLLABUS MAV-12-04

- Solve problems involving equations of tangents and normals to curves involving any of the functions within the scope of the Mathematics Advanced course

NOTES

- The gradient of the tangent at $x = a$ is $f'(a)$.
- The normal is perpendicular to the tangent, so its gradient is $-\dfrac{1}{f'(a)}$.
- Find the point with the original function: $y = f(a)$. Find the gradient with the derivative.
- Use $y - y_1 = m(x - x_1)$, then rearrange to $y = mx + c$ or general form.
- Leave e in exact answers: $e^0 = 1$, $e^1 = e$, $e^{\ln k} = k$.



WORKED EXAMPLE

Find the equation of the tangent to $y = e^{4x}$ at the point where $x = 1$.

$x = 1$: $y = e^4$, so the point is $(1, e^4)$. — point from f

$\dfrac{dy}{dx} = 4e^{4x}$, so $m = 4e^4$. — gradient from f'

$y - e^4 = 4e^4(x - 1)$ — point–gradient form

$y = 4e^4x - 3e^4$

Tier 1 · tangents to $y = e^{kx}$

58. Find the exact value.

a) e^{4x} at $x = 1$

b) $3e^{1-x}$ at $x = 1$

c) xe^{3x} at $x = 1$

59. The tangent has gradient m. Find the gradient of the normal.

a) $m = 5e$

b) $m = e^4$

c) $m = -\dfrac{3}{e}$

60. Find the equation of the tangent to $y = e^x$ at the point where $x = 2$.

61. Find the equation of the tangent to $y = e^{5x}$ at the point where $x = 0$.

62. Find the equation of the tangent to $y = 2e^{3x}$ at the point where $x = 1$.

Tier 2 · normals

63. Find the equation of the normal to $y = e^{3x}$ at the point where $x = 0$. Give your answer in general form.

64. Find the equation of the normal to $y = e^x$ at the point where $x = 1$. Give your answer in general form.

65. Find the equation of the normal to $y = e^{-2x}$ at the point where $x = 0$. Give your answer in general form.

Tier 3 · tangents that need the product rule

66. Find the equation of the tangent to $y = xe^{2x}$ at the origin.

67. Find the equation of the tangent to $y = xe^{-x}$ at the point where $x = 2$.

68. Find the equation of the tangent to $y = x^2e^x$ at the point where $x = 1$.

Tier 4 · normals in general form

69. Find the equation of the normal to $y = e^{x^2}$ at the point where $x = -1$. Give your answer in general form.

70. Find the equation of the normal to $y = e^{2x-2}$ at the point where $x = 1$. Give your answer in general form.

71. Find the equation of the normal to $y = e^{x^2-4}$ at the point where $x = 2$. Give your answer in general form.

Tier 5 · show that

72. Show that the tangent to $y = e^{2x}$ at the point where $x = a$ meets the x-axis at $x = a - \frac{1}{2}$.

73. Show that the tangent to $y = e^{-x}$ at the point where $x = a$ meets the x-axis at $x = a + 1$.

74. Show that the tangent to $y = e^x$ at the point where $x = a$ meets the y-axis at $y = (1 - a)e^a$.

Tier 6 · finding the point

75. The tangent to $y = e^{2x}$ at the point where $x = a$ passes through the origin. Find a and the equation of the tangent.

76. Find the exact coordinates of the point on $y = e^x$ where the tangent has gradient 3.

77. Find the exact coordinates of the point on $y = e^{3x}$ where the normal has gradient $-\frac{1}{6}$.

Conceptual

78. Explain why no tangent to $y = e^x$ is horizontal.

79. Show that the tangents to $y = e^x$ and $y = e^{-x}$ at the point $(0, 1)$ are perpendicular.

80. Find where the normal to $y = e^x$ at the point where $x = 0$ crosses the x-axis.

81. A student finds the tangent to $y = e^{2x}$ at $x = 1$ using the point $(1, 2e^2)$. Explain the error.

82. Let $f(x) = e^{2x}$.

(a) Find the equation of the tangent to $y = f(x)$ at the point where $x = 0$. (2 marks)

(b) Find where this tangent meets the x-axis. (1 mark)

Revision

83. Let $f(x) = xe^{-2x}$.

(a) Show that $f'(x) = e^{-2x}(1 - 2x)$. (2 marks)

(b) Find the exact coordinates of the point where the tangent to $y = f(x)$ is horizontal. (2 marks)

(c) For what values of x is $f(x)$ increasing? (1 mark)

84. The line AB has equation $y = 3x - 3$. It meets the y-axis at A and the x-axis at B. The line BC is perpendicular to AB, and C lies on the y-axis.

(a) Show that the length of AB is $\sqrt{10}$. (1 mark)

(b) Find the equation of the line BC. (2 marks)

(c) Find the area of triangle ABC. (2 marks)

85.

(a) Solve $3^{2x+1} = 81$. (1 mark)

(b) Solve $2^{2x} - 3(2^x) - 4 = 0$. (3 marks)

(c) Solve $2^x = 10$. Give your answer correct to three significant figures. (1 mark)

4 The derivative of $\ln x$ and $\ln f(x)$

ANCHOR QUESTION

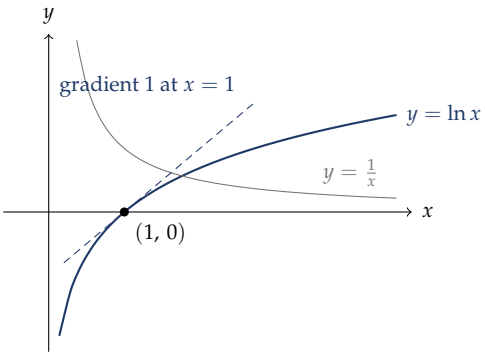
Differentiate $y = \ln\left(x^2 - 3x + 1\right)$.

SYLLABUS MAV-12-04

- Use chain rule on the identity $e^{\ln x} = x$ to prove the formula $\dfrac{d}{dx}(\ln x) = \dfrac{1}{x}$ for the derivative of the natural logarithm function where $x > 0$
- Use the chain rule to differentiate functions of the form $\ln f(x)$

NOTES

- $\dfrac{d}{dx}(\ln x) = \dfrac{1}{x}$, for $x > 0$.
- Proof: differentiate both sides of $e^{\ln x} = x$. Then $e^{\ln x} \times \dfrac{d}{dx}(\ln x) = 1$, so $x\,\dfrac{d}{dx}(\ln x) = 1$.
- $\dfrac{d}{dx}\big(\ln f(x)\big) = \dfrac{f'(x)}{f(x)}$: the derivative of the inside, over the inside. This is on the reference sheet.
- $\ln(kx) = \ln k + \ln x$, so $\ln(kx)$ has the same derivative as $\ln x$: $\dfrac{1}{x}$.
- A product, quotient or power inside the log: expand with the log laws first. $\ln(AB) = \ln A + \ln B$, $\ln\dfrac{A}{B} = \ln A - \ln B$, $\ln A^n = n \ln A$.
- A sum or difference inside the log: differentiate it as it is.



WORKED EXAMPLE

Differentiate $y = \ln\left(x^2 + 4x - 1\right)$.

$f(x) = x^2 + 4x - 1$, so $f'(x) = 2x + 4$. the inside and its derivative

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)} = \frac{2x+4}{x^2+4x-1}$$

Tier 1 · the log of a polynomial

86. For each function, write the inside $f(x)$, then find $f'(x)$.

a) $\ln(9x)$

b) $\ln\left(x^2-7\right)$

c) $\ln(5-2x)$

87. Expand using the log laws.

a) $\ln\left(x^3(x+2)\right)$

b) $\ln\left(\dfrac{x-1}{x+5}\right)$

c) $\ln\left((4x+3)^2\right)$

88. Should you use the log laws first? Answer yes or no.

a) $\ln\left((x+1)(x+2)\right)$

b) $\ln\left(2x^3+1\right)$

c) $\ln\left((x-3)^4\right)$

89. Differentiate:

a) $y=\ln\left(x^2+1\right)$

b) $y=\ln\left(x^3-4x\right)$

90. Differentiate $y=\ln\left(2x^2-5x+3\right)$.

Tier 2 · a linear inside

91. Differentiate $y=\ln(6x)$.

92. Differentiate $y=\ln(3x+2)$.

93. Differentiate $y=\ln(1-4x)$.

Tier 3 · coefficients and sums

94. Differentiate $y=5\ln(2x+3)$.

95. Differentiate $y=x^3-2\ln x$.

96. Differentiate $y=\ln\left(x^2+1\right)+e^{2x}$.

Tier 4 · the power law first

97. Differentiate $y = \ln\left((2x+1)^3\right)$.

98. Differentiate $y = \ln\sqrt{4x-1}$.

99. Differentiate $y = \ln\left(x^5\right)$.

Tier 5 · the product and quotient laws first

100. Differentiate $y = \ln\big(x(x+3)\big)$, giving your answer as a single fraction.

101. Differentiate $y = \ln\left(\dfrac{x+2}{x-1}\right)$, giving your answer as a single fraction.

102. Differentiate $y = \ln\left(\dfrac{3x-1}{x+1}\right)$, giving your answer as a single fraction.

Tier 6 · several log laws

103. Differentiate $y = \ln\left(x^3\sqrt{x+1}\right)$, giving your answer as a single fraction.

104. Differentiate $y = \ln\left(\dfrac{(x+1)^2}{x-2}\right)$, giving your answer as a single fraction.

105. Differentiate $y = \ln\sqrt{\dfrac{x+1}{x-1}}$, simplifying fully.

Conceptual

106. Use the identity $e^{\ln x} = x$ to prove that $\dfrac{d}{dx}(\ln x) = \dfrac{1}{x}$.

107. Show that $\ln(5x)$ and $\ln x$ have the same derivative, and explain why using the log laws.

108. A student writes $\dfrac{d}{dx}\ln\left(x^2+1\right) = \dfrac{1}{x^2+1}$. Explain the error and give the correct derivative.

109. Find the exact gradient of the tangent to $y = \ln\left(x^2+3\right)$ at the point where $x = 1$.

110. Let $f(x) = \ln\left(x^2 - 2x + 5\right)$.

(a) Find $f'(x)$. (1 mark)

(b) Explain why $f(x)$ is defined for all real x. (1 mark)

(c) Find the exact coordinates of the point where the tangent to $y = f(x)$ is horizontal. (1 mark)

Revision

111. Consider the curve $y = e^x$.

(a) Find the equation of the tangent to the curve at the point where $x = 1$. (2 marks)

(b) Show that this tangent passes through the origin. (1 mark)

(c) Find the equation of the normal to the curve at the point where $x = 0$, and the point where it meets the x-axis. (2 marks)

112. Let $f(x) = x^3 + 3x^2 + kx - 1$, where k is a constant.

(a) Find $f'(x)$. (1 mark)

(b) Find the values of k for which $f(x)$ is an increasing function. (2 marks)

(c) When $k = -9$, find the x-coordinates of the points where the tangent to $y = f(x)$ is horizontal. (2 marks)

113. James parks his car on one of four levels of a car park, chosen at random. When he leaves, he chooses a level at random and searches it. If his car is not there, he chooses a different level, and continues until he finds it.

(a) What is the probability that his car is on the first level he searches? (1 mark)

(b) What is the probability that he must search all four levels? (1 mark)

(c) What is the probability that, on each of the five working days in a week, his car is not on the first level he searches? (1 mark)

(d) What is the probability that, on at least one of those five days, his car is on the first level he searches? (1 mark)

5 $\log_a x$, combined rules and tangents

ANCHOR QUESTION

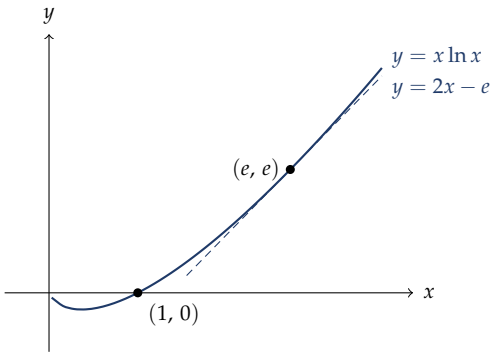
Differentiate $y = x^2 \ln x$, giving your answer in factorised form.

SYLLABUS MAV-12-04

- Prove and use the formula $\dfrac{d}{dx}(\log_a x) = \dfrac{1}{x \ln a}$, where a is a constant and $a > 0$
- Apply the product, quotient and chain rules to differentiate functions of the form $f(x)g(x)$, $\dfrac{f(x)}{g(x)}$ or $f(g(x))$, where $f(x)$ and $g(x)$ are any of the functions within the scope of the Mathematics Advanced course
- Solve problems involving equations of tangents and normals to curves involving any of the functions within the scope of the Mathematics Advanced course

NOTES

- Change of base: $\log_a x = \dfrac{\ln x}{\ln a}$.
- $\dfrac{d}{dx}(\log_a x) = \dfrac{1}{x \ln a}$. Proof: $\log_a x = \dfrac{1}{\ln a} \cdot \ln x$, and $\dfrac{1}{\ln a}$ is a constant.
- For $\log_a f(x)$: change to $\dfrac{\ln f(x)}{\ln a}$, then use $\dfrac{f'(x)}{f(x)}$.
- A power inside the log, as in $\ln(x^3)$: log law first, $3 \ln x$.
- A power on the whole log, as in $(\ln x)^3$: chain rule on the outside, $3(\ln x)^2 \times \dfrac{1}{x}$. Note $\ln^n x$ means $(\ln x)^n$.
- Tangents and normals at $x = e$: use $\ln e = 1$ and $\ln 1 = 0$ to keep answers exact.



WORKED EXAMPLE

Differentiate $y = x^4 \ln x$, giving your answer in factorised form.

$u = x^4$, $u' = 4x^3$; $\quad v = \ln x$, $v' = \dfrac{1}{x}$ — name the parts

$\dfrac{dy}{dx} = 4x^3 \ln x + x^4 \times \dfrac{1}{x} = 4x^3 \ln x + x^3$ — product rule

$= x^3(4 \ln x + 1)$ — common factor x^3

Tier 1 · products with $\ln x$

114. Write using natural logarithms.

a) $\log_6 x$
b) $\log_3 (x^2 - 1)$

115. Differentiate:

a) $y = \log_4 x$
b) $y = 3\log_5 x$
c) $y = \log_2(3x - 1)$

116. For each function, state whether to use a log law first or the chain rule on the outside.

a) $\ln (x^6)$
b) $(\ln x)^6$
c) $\ln^2(3x)$
d) $\ln ((x+4)^3)$

117. Differentiate $y = x \ln x$.

118. Differentiate $y = x^5 \ln x$, giving your answer in factorised form.

Tier 2 · other products with logs

119. Differentiate $y = e^x \ln x$.

120. Differentiate $y = x^2 \ln(3x)$, giving your answer in factorised form.

121. Differentiate $y = \sqrt{x}\, \ln x$, giving your answer as a single fraction.

Tier 3 · the quotient rule with logs

122. Differentiate $y = \dfrac{\ln x}{x^2}$.

123. Differentiate $y = \dfrac{x}{\ln x}$.

124. Differentiate $y = \dfrac{\ln x}{x+1}$, giving your answer as a single fraction.

Tier 4 · powers of $\ln x$

125. Differentiate $y = (\ln x)^2$.

126. Differentiate $y = (\ln x)^4$.

127. Differentiate $y = \sqrt{\ln x}$.

Tier 5 · tangents with logs

128. Find the equation of the tangent to $y = \ln x$ at the point where $x = 1$.

129. Find the equation of the tangent to $y = \ln(3x)$ at the point where $x = \frac{1}{3}$.

130. Find the equation of the tangent to $y = 2\ln x$ at the point where $x = e$.

Tier 6 · normals and combined rules

131. Find the equation of the normal to $y = \ln x$ at the point where $x = e$.

132. Find the equation of the tangent to $y = x^2 \ln x$ at the point where $x = 1$.

133. Find the equation of the normal to $y = x\ln x$ at the point where $x = 1$.

Conceptual

134. Prove that $\dfrac{d}{dx}(\log_a x) = \dfrac{1}{x\ln a}$.

135. Find $\dfrac{d}{dx}\ln\left(x^2\right)$ and $\dfrac{d}{dx}(\ln x)^2$, and explain why they are different.

136. Show that the tangent to $y = \ln x$ at the point where $x = e$ passes through the origin.

137. Find the exact coordinates of the point on $y = x\ln x$ where the tangent is horizontal.

138. Let $f(x) = \dfrac{\ln x}{x}$, for $x > 0$.

(a) Show that $f'(x) = \dfrac{1 - \ln x}{x^2}$. (1 mark)

(b) Find the exact coordinates of the point where the tangent to $y = f(x)$ is horizontal. (2 marks)

Revision

139. Let $f(x) = \ln\left(\dfrac{x+1}{x-1}\right)$, for $x > 1$.

(a) Show that $f'(x) = -\dfrac{2}{x^2-1}$. (2 marks)

(b) Explain why $f(x)$ is decreasing for $x > 1$. (1 mark)

(c) Find the equation of the tangent to $y = f(x)$ at the point where $x = 3$. (2 marks)

140.

(a) Solve $e^{2x} - 3e^x + 2 = 0$. (3 marks)

(b) Solve $\log_2 x + \dfrac{3}{\log_2 x} = 4$. (3 marks)

141. The graph of $y = A\sin bx$, where $A > 0$ and $b > 0$, has amplitude 3 and period π.

(a) Find the values of A and b. (2 marks)

(b) Find the exact values of x, for $0 \leq x \leq \pi$, where $y = \dfrac{3\sqrt{3}}{2}$. (2 marks)

6 Derivatives of $\sin x$, $\cos x$ and $\tan x$

ANCHOR QUESTION

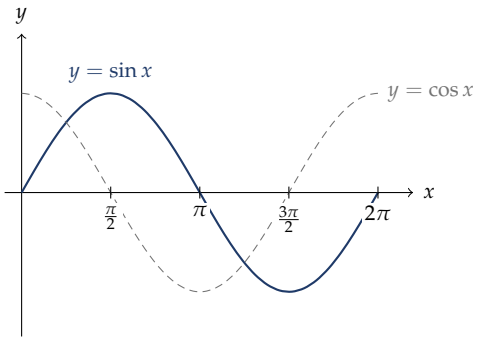
Differentiate $y = \sin\left(x^2 + 2x\right)$.

SYLLABUS MAV-12-04

- Determine the formulas $\frac{d}{dx}(\sin x) = \cos x$ and $\frac{d}{dx}(\cos x) = -\sin x$ informally by using the graphs of $y = \sin x$ and $y = \cos x$ and verify with graphing applications
- Use the rules of differentiation to show that $\frac{d}{dx}(\tan x) = \sec^2 x$
- Use the chain rule to differentiate functions of the form $\sin f(x)$, $\cos f(x)$ and $\tan f(x)$

NOTES

- $\frac{d}{dx}(\sin x) = \cos x$ and $\frac{d}{dx}(\cos x) = -\sin x$. Angles must be in radians.
- From the graph of $y = \sin x$: the gradient is 0 at the turning points $x = \frac{\pi}{2}$, $\frac{3\pi}{2}$, is 1 at $x = 0$ and -1 at $x = \pi$. That is the graph of $y = \cos x$.
- $\frac{d}{dx}(\tan x) = \sec^2 x$, where $\sec x = \frac{1}{\cos x}$. Proof: write $\tan x = \frac{\sin x}{\cos x}$ and use the quotient rule with $\sin^2 x + \cos^2 x = 1$.
- Chain rule: $\frac{d}{dx}\sin f(x) = f'(x)\cos f(x)$, $\frac{d}{dx}\cos f(x) = -f'(x)\sin f(x)$, $\frac{d}{dx}\tan f(x) = f'(x)\sec^2 f(x)$.
- The angle stays the same; multiply by the derivative of the angle.



WORKED EXAMPLE

Differentiate $y = \cos\left(x^2 + 4x\right)$.

$f(x) = x^2 + 4x$, so $f'(x) = 2x + 4$. — the angle and its derivative

$\frac{dy}{dx} = -f'(x)\sin f(x)$ — cos gives − sin

$= -(2x + 4)\sin\left(x^2 + 4x\right)$

Tier 1 · $\sin f(x)$

142. For each function, write the angle $f(x)$, then find $f'(x)$.

a) $\sin(9x)$

b) $\cos\left(x^2-1\right)$

c) $\tan(5-2x)$

143. Differentiate:

a) $y=\sin x+\cos x$

b) $y=4\tan x$

c) $y=x^2-3\cos x$

144. Differentiate $y=\sin\left(x^2+3x\right)$.

145. Differentiate $y=\sin\left(x^3\right)$.

146. Differentiate $y=\sin(5x-2)$.

Tier 2 · $\cos f(x)$

147. Differentiate $y=\cos\left(x^2+1\right)$.

148. Differentiate $y=\cos 4x$.

149. Differentiate $y=\cos(2-3x)$.

Tier 3 · $\tan f(x)$

150. Differentiate $y=\tan 6x$.

151. Differentiate $y=\tan\left(x^2\right)$.

152. Differentiate $y=\tan(1-x)$.

Tier 4 · coefficients and sums

153. Differentiate $y=5\sin 3x+2\cos 4x$.

154. Differentiate $y=2\tan\dfrac{x}{2}$.

155. Differentiate $y=\cos(\pi x)-\sin(\pi x)$.

Tier 5 · exact gradients

156. Find the exact gradient of the tangent to $y = \cos 2x$ at the point where $x = \dfrac{\pi}{12}$.

157. Find the exact gradient of the tangent to $y = \tan x$ at the point where $x = \dfrac{\pi}{4}$.

158. Find the exact gradient of the tangent to $y = \sin 3x$ at the point where $x = \dfrac{\pi}{9}$.

Tier 6 · horizontal tangents

159. Find the values of x, for $0 \leq x \leq 2\pi$, where the tangent to $y = \sin x - \cos x$ is horizontal.

160. Find the values of x, for $0 \leq x \leq 2\pi$, where the tangent to $y = x + 2\cos x$ is horizontal.

161. Find the values of x, for $0 \leq x \leq \pi$, where the tangent to $y = \sin 2x$ is horizontal.

Conceptual

162. Use the quotient rule to show that $\dfrac{d}{dx}(\tan x) = \sec^2 x$.

163. Use the graph of $y = \sin x$ to explain why its gradient is 1 at $x = 0$, 0 at $x = \dfrac{\pi}{2}$ and -1 at $x = \pi$, and why this matches $y = \cos x$.

164. A student writes $\dfrac{d}{dx}(\cos 3x) = -\sin 3x$. Explain the error and give the correct derivative.

165. Explain why the graph of $y = \tan x$ has no horizontal tangents.

166. Let $f(x) = \sin x + \sqrt{3}\cos x$, for $0 \leq x \leq 2\pi$.

(a) Find $f'(x)$. (1 mark)

(b) Find the values of x where the tangent to $y = f(x)$ is horizontal. (2 marks)

Revision

167. Let $f(x) = x^3 \ln x$, for $x > 0$.

(a) Show that $f'(x) = x^2(3 \ln x + 1)$. (2 marks)

(b) Find the exact value of x where the tangent to $y = f(x)$ is horizontal. (2 marks)

(c) Find the equation of the tangent to $y = f(x)$ at the point where $x = e$. (2 marks)

168.

(a) Prove that $\sec^2 x + \sec x \tan x = \dfrac{1 + \sin x}{\cos^2 x}$. (1 mark)

(b) Hence prove that $\sec^2 x + \sec x \tan x = \dfrac{1}{1 - \sin x}$. (1 mark)

(c) Hence solve $\sec^2 x + \sec x \tan x = 2$ for $0 \leq x \leq 2\pi$. (2 marks)

169. The points A and B lie on a circle with centre O and radius 3 cm, and $\angle AOB = \dfrac{\pi}{3}$.

(a) Find the exact area of the sector AOB. (1 mark)

(b) Find the exact perimeter of the minor segment bounded by the chord AB and the arc AB. (2 marks)

(c) Find the exact area of the minor segment. (2 marks)

7 Trig functions with combined rules

ANCHOR QUESTION

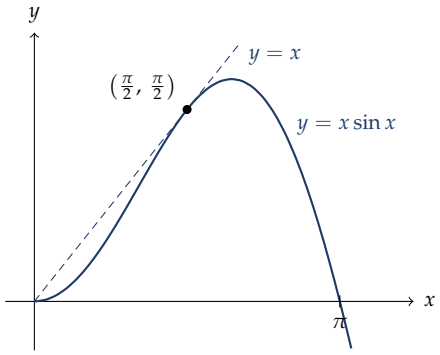
Differentiate $y = x^2 \sin x$, giving your answer in factorised form.

SYLLABUS MAV-12-04

- Apply the product, quotient and chain rules to differentiate functions of the form $f(x)g(x)$, $\dfrac{f(x)}{g(x)}$ or $f(g(x))$, where $f(x)$ and $g(x)$ are any of the functions within the scope of the Mathematics Advanced course
- Solve problems involving equations of tangents and normals to curves involving any of the functions within the scope of the Mathematics Advanced course

NOTES

- $\sin^n x$ means $(\sin x)^n$: use the chain rule on the outside, e.g. $\dfrac{d}{dx}\sin^2 x = 2\sin x \cos x$.
- Product rule $u'v + uv'$ and quotient rule $\dfrac{u'v - uv'}{v^2}$ work with trig functions exactly as before. Name u, v, u', v' first.
- Take out common factors, e.g. $2x\sin x + x^2\cos x = x(2\sin x + x\cos x)$.
- Trig inside e or ln: $\dfrac{d}{dx}e^{\sin x} = \cos x\, e^{\sin x}$ and $\dfrac{d}{dx}\ln(\cos x) = \dfrac{-\sin x}{\cos x} = -\tan x$.
- For tangents, use exact values: $\sin\dfrac{\pi}{2} = 1$, $\cos\dfrac{\pi}{2} = 0$, $\cos\pi = -1$.



WORKED EXAMPLE

Differentiate $y = x^3\cos x$, giving your answer in factorised form.

$u = x^3$, $u' = 3x^2$; $\quad v = \cos x$, $v' = -\sin x$ — name the parts

$\dfrac{dy}{dx} = 3x^2\cos x - x^3\sin x$ — product rule

$= x^2(3\cos x - x\sin x)$ — common factor x^2

Tier 1 · products with $\sin x$, $\cos x$, $\tan x$

170. Rewrite with the power outside a bracket.

a) $\cos^3 x$

b) $\sin^2 4x$

171. Differentiate $y = x \sin x$.

172. Differentiate $y = x^2 \cos x$, giving your answer in factorised form.

173. Differentiate $y = x^3 \tan x$, giving your answer in factorised form.

174. Differentiate $y = \sqrt{x} \sin x$.

Tier 2 · products with a chain inside

175. Differentiate $y = x \sin 3x$.

176. Differentiate $y = x^2 \cos 2x$, giving your answer in factorised form.

177. Differentiate $y = \sin x \cos x$.

Tier 3 · powers of trig functions

178. Differentiate $y = \cos^2 x$.

179. Differentiate $y = \sin^3 x$.

180. Differentiate $y = \tan^2 x$.

Tier 4 · the quotient rule with trig

181. Differentiate $y = \dfrac{\cos x}{x}$.

182. Differentiate $y = \dfrac{x}{\sin x}$.

183. Show that $\dfrac{d}{dx}\left(\dfrac{\sin x}{1+\cos x}\right) = \dfrac{1}{1+\cos x}$.

Tier 5 · trig with e and $\ln$

184. Differentiate $y = e^x \sin x$, giving your answer in factorised form.

185. Differentiate $y = \ln(\sin x)$.

186. Differentiate $y = e^{\cos x}$.

Tier 6 · tangents and normals

187. Find the equation of the tangent to $y = x\cos x$ at the point where $x = \pi$.

188. Find the equation of the tangent to $y = \sin^2 x$ at the point where $x = \dfrac{\pi}{4}$.

189. Find the equation of the normal to $y = e^x \cos x$ at the point where $x = 0$.

Tier 7 · show that

190. Show that $\dfrac{d}{dx}\left(\dfrac{\cos x}{1-\sin x}\right) = \dfrac{1}{1-\sin x}$.

191. Show that $\dfrac{d}{dx}(x\sin x + \cos x) = x\cos x$.

192. Show that $\dfrac{d}{dx}\left[e^{-x}(\sin x + \cos x)\right] = -2e^{-x}\sin x$.

Conceptual

193. Show that $\dfrac{d}{dx}\left(\sin^2 x + \cos^2 x\right) = 0$, and explain why this makes sense.

194. Show that $\dfrac{d}{dx}\ln(\sin x) = \cot x$.

195. A student writes $\dfrac{d}{dx}(x\sin x) = \cos x$. Explain the error and give the correct derivative.

196. Find the exact gradient of the tangent to $y = x^2\sin x$ at the point where $x = \pi$.

197. Let $f(x) = e^{-x}\sin x$, for $0 \le x \le 2\pi$.

(a) Show that $f'(x) = e^{-x}(\cos x - \sin x)$. (1 mark)

(b) Find the values of x where the tangent to $y = f(x)$ is horizontal. (2 marks)

Revision

198. Consider the curve $y = 2\sin x - x$ for $0 \leq x \leq 2\pi$.

(a) Find $\dfrac{dy}{dx}$. (1 mark)

(b) Find the values of x where the tangent to the curve is horizontal. (2 marks)

(c) Find the equation of the tangent to the curve at the point where $x = \pi$. (2 marks)

199. Let $f(x) = \dfrac{\ln x}{x}$, for $x > 0$.

(a) Show that $f'(x) = \dfrac{1 - \ln x}{x^2}$. (2 marks)

(b) Find the exact coordinates of the point where the tangent to $y = f(x)$ is horizontal. (2 marks)

(c) Find the equation of the tangent to $y = f(x)$ at the point where $x = 1$. (2 marks)

200.

(a) Prove that $1 + \tan^2 x = \sec^2 x$. (1 mark)

(b) Hence solve $\tan^2 x = 2\sec x - 1$ for $0 \leq x \leq 2\pi$. (3 marks)

8 Derivatives of the reciprocal trig functions

ANCHOR QUESTION

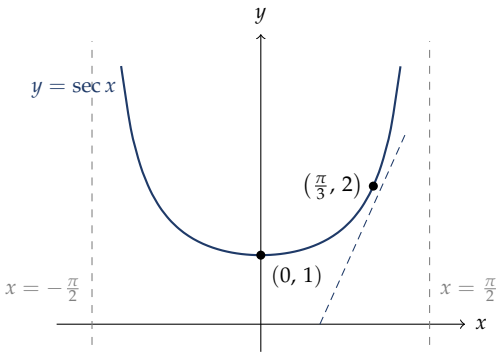
Show that $\dfrac{d}{dx}(\sec x) = \tan x \sec x$.

SYLLABUS MAV-12-04

- Use the rules of differentiation to find the derivatives of cosec x, sec x and cot x
- Solve problems involving equations of tangents and normals to curves involving any of the functions within the scope of the Mathematics Advanced course

NOTES

- $\sec x = \dfrac{1}{\cos x} = (\cos x)^{-1}$, $\operatorname{cosec} x = \dfrac{1}{\sin x} = (\sin x)^{-1}$, $\cot x = \dfrac{\cos x}{\sin x}$.
- $\dfrac{d}{dx}(\sec x) = \tan x \sec x$: chain rule on $(\cos x)^{-1}$.
- $\dfrac{d}{dx}(\operatorname{cosec} x) = -\cot x \operatorname{cosec} x$: chain rule on $(\sin x)^{-1}$.
- $\dfrac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$: quotient rule on $\dfrac{\cos x}{\sin x}$.
- These three are *not* on the reference sheet. The "co" functions get the minus sign.
- With an angle $f(x)$, multiply by $f'(x)$ as usual, e.g. $\dfrac{d}{dx}\sec 2x = 2\tan 2x \sec 2x$.



WORKED EXAMPLE

Show that $\dfrac{d}{dx}(\sec 4x) = 4\tan 4x \sec 4x$.

$y = (\cos 4x)^{-1}$ — write as a power

$\dfrac{dy}{dx} = -(\cos 4x)^{-2} \times (-4\sin 4x)$ — chain rule

$= \dfrac{4\sin 4x}{\cos^2 4x} = 4 \cdot \dfrac{\sin 4x}{\cos 4x} \cdot \dfrac{1}{\cos 4x} = 4\tan 4x \sec 4x$

Tier 1 · $\sec x$ as a power

201. Write as a power of a trig function.

a) $\operatorname{cosec} 5x$

b) $\sec\left(x^3\right)$

202. Find the exact value.

a) $\sec \dfrac{\pi}{6}$

b) $\operatorname{cosec} \dfrac{\pi}{3}$

c) $\cot \dfrac{\pi}{4}$

d) $\operatorname{cosec} \dfrac{\pi}{2}$

203. By writing $\sec 3x = (\cos 3x)^{-1}$, show that $\dfrac{d}{dx}(\sec 3x) = 3\tan 3x \sec 3x$.

204. Differentiate $y = \sec^2 x$.

205. Differentiate $y = 5\sec x$.

Tier 2 · $\operatorname{cosec} x$

206. Differentiate $y = \operatorname{cosec} x + x^2$.

207. Differentiate $y = 2\operatorname{cosec} x$.

208. Show that $\dfrac{d}{dx}(\operatorname{cosec} 2x) = -2\cot 2x \operatorname{cosec} 2x$.

Tier 3 · $\cot x$

209. Differentiate $y = \cot x - x$.

210. Differentiate $y = 4\cot x$.

211. By writing $\cot x = (\tan x)^{-1}$, show that $\dfrac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$.

Tier 4 · sums

212. Differentiate $y = \sec x + \operatorname{cosec} x$.

213. Differentiate $y = 2\cot x - 3\sec x$.

214. Differentiate $y = \tan x + \cot x$.

Tier 5 · the chain rule

215. Differentiate $y = \sec\left(x^2\right)$.

216. Differentiate $y = \cot 3x$.

217. Differentiate $y = \operatorname{cosec}(2x - 1)$.

Tier 6 · the product rule

218. Differentiate $y = x \operatorname{cosec} x$, giving your answer in factorised form.

219. Differentiate $y = x^2 \cot x$, giving your answer in factorised form.

220. Differentiate $y = e^x \sec x$, giving your answer in factorised form.

Tier 7 · tangents and normals

221. Find the equation of the tangent to $y = \sec x$ at the point where $x = \dfrac{\pi}{3}$.

222. Find the gradient of the normal to $y = \operatorname{cosec} x$ at the point where $x = \dfrac{\pi}{4}$.

223. Find the equation of the tangent to $y = \cot x$ at the point where $x = \dfrac{\pi}{2}$.

Conceptual

224. Show that $\dfrac{d}{dx}(\sec x) = \tan x \sec x$.

225. Use the quotient rule to show that $\dfrac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$.

226. Explain why $y = \cot x$ is decreasing wherever it is defined.

227. A student says that since $\sec x = \dfrac{1}{\cos x}$, its derivative is $\dfrac{1}{-\sin x}$. Explain the error.

228. Let $f(x) = \sec x$, for $-\dfrac{\pi}{2} < x < \dfrac{\pi}{2}$.

(a) Show that $f'(x) = \dfrac{\sin x}{\cos^2 x}$. (1 mark)

(b) Find the point where the tangent to $y = f(x)$ is horizontal. (1 mark)

(c) Explain why $f'(x) > 0$ for $0 < x < \dfrac{\pi}{2}$. (1 mark)

Revision

229. Consider the curve $y = \log_e(\sin x)$ for $0 < x < \pi$.

(a) Show that $\dfrac{dy}{dx} = \dfrac{\cos x}{\sin x}$. (2 marks)

(b) Find the equation of the tangent to the curve at the point where $x = \dfrac{\pi}{4}$. (3 marks)

230. Let $f(x) = (x+2)(x^2+4)$.

(a) Show that the graph of $y = f(x)$ has no horizontal tangents. (2 marks)

(b) Find the points where the graph crosses the axes. (1 mark)

(c) Find the equation of the tangent to $y = f(x)$ at the point where $x = 0$. (2 marks)

231. The line $y = 2x + c$ is a tangent to the parabola $y = x^2 + 3$.

(a) Find the value of c. (2 marks)

(b) Find the point of contact. (1 mark)

(c) The normal to the parabola at the point of contact meets the parabola again at Q. Find the coordinates of Q. (2 marks)

9 Mixed differentiation: choosing the rule

ANCHOR QUESTION

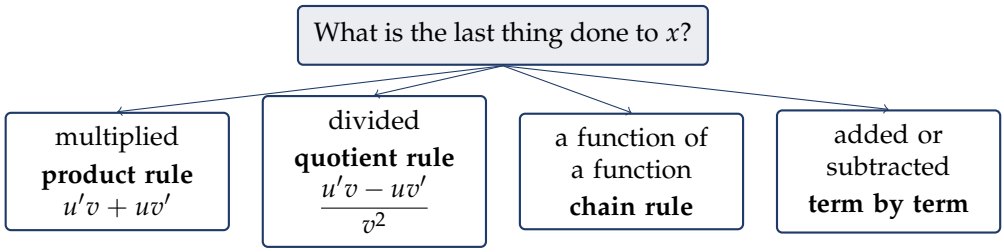
Differentiate $y = e^{2x}\sin 3x$, giving your answer in factorised form.

SYLLABUS MAV-12-04

- Apply the product, quotient and chain rules to differentiate functions of the form $f(x)g(x)$, $\dfrac{f(x)}{g(x)}$ or $f(g(x))$, where $f(x)$ and $g(x)$ are any of the functions within the scope of the Mathematics Advanced course
- Solve problems involving equations of tangents and normals to curves involving any of the functions within the scope of the Mathematics Advanced course

NOTES

- Standard derivatives: $e^x \to e^x$, $\ln x \to \dfrac{1}{x}$, $a^x \to (\ln a)\,a^x$, $\sin x \to \cos x$, $\cos x \to -\sin x$, $\tan x \to \sec^2 x$, $\sec x \to \tan x \sec x$, $\operatorname{cosec} x \to -\cot x \operatorname{cosec} x$, $\cot x \to -\operatorname{cosec}^2 x$.
- Choose the rule from the *last* thing done to x: multiplied, divided, a function of a function, or added.
- Inside a product or quotient, each part may need its own chain rule, e.g. $\dfrac{d}{dx}\sin 3x = 3\cos 3x$.
- Simplify by taking out common factors, especially exponentials.
- $e^{f(x)}$, $\ln f(x)$, $\sin f(x)$ and $[f(x)]^n$ all follow the chain rule: keep the inside, multiply by its derivative.



WORKED EXAMPLE

Differentiate $y = e^{-x}\sin 2x$, giving your answer in factorised form.

Last operation: multiplied, so the product rule. — choose the rule

$u = e^{-x}$, $u' = -e^{-x}$; $\quad v = \sin 2x$, $v' = 2\cos 2x$ — chain rule on each part

$$\frac{dy}{dx} = -e^{-x}\sin 2x + 2e^{-x}\cos 2x$$

$$= e^{-x}(2\cos 2x - \sin 2x)$$ — common factor e^{-x}

Tier 1 · products with a chain inside

232. Differentiate:

a) e^x

b) $\ln x$

c) $\sin x$

d) $\cot x$

e) 3^x

233. Which rule comes first?

a) $x^2 \cos x$

b) $\cos\left(x^2\right)$

c) $\dfrac{\cos x}{x^2}$

d) $x^2 + \cos x$

234. Differentiate $y = e^x \cos 2x$, giving your answer in factorised form.

235. Differentiate $y = x^2 e^{3x}$, giving your answer in factorised form.

Tier 2 · products with a negative power

236. Differentiate $y = x^2 e^{-x}$, giving your answer in factorised form.

237. Differentiate $y = xe^{-3x}$, giving your answer in factorised form.

238. Differentiate $y = e^{-x} \cos x$, giving your answer in factorised form.

Tier 3 · quotients

239. Differentiate $y = \dfrac{\ln x}{e^x}$, giving your answer as a single fraction.

240. Differentiate $y = \dfrac{\sin x}{e^{2x}}$.

241. Differentiate $y = \dfrac{x^2}{\ln x}$, giving your answer in factorised form.

Tier 4 · a function of a function

242. Differentiate $y = \cos\left(e^x\right)$.

243. Differentiate $y = e^{\tan x}$.

244. Differentiate $y = \ln\left(1 + e^x\right)$.

Tier 5 · the log of a sum

245. Differentiate $y = \ln(x + \cos x)$.

246. Differentiate $y = \ln\left(e^{x} + e^{-x}\right)$.

247. Differentiate $y = \ln(2x + \sin 2x)$.

Tier 6 · powers of functions

248. Differentiate $y = (e^{x} + 1)^{3}$.

249. Differentiate $y = (\sin x + \cos x)^{2}$.

250. Differentiate $y = \sqrt{1 + \ln x}$.

Tier 7 · quotients with a chain

251. Show that $\dfrac{d}{dx}\left(\dfrac{\sin x}{(1+\cos x)^{2}}\right) = \dfrac{2 - \cos x}{(1+\cos x)^{2}}$.

252. Differentiate $y = \dfrac{x}{\left(x^{2}+1\right)^{2}}$, simplifying fully.

253. Differentiate $y = \dfrac{e^{x}}{\left(e^{x}+1\right)^{3}}$, simplifying fully.

Tier 8 · tangents and normals

254. Find the equation of the tangent to $y = e^{x}\sin x$ at the origin.

255. Find the equation of the tangent to $y = \ln\left(1 + x^{2}\right)$ at the point where $x = 1$.

256. Find the equation of the normal to $y = xe^{2x}$ at the origin.

Conceptual

257. Explain why $y = e^{x}\sin x$ needs the product rule but $y = e^{\sin x}$ needs the chain rule, and differentiate both.

258. Show that $\dfrac{d}{dx}\ln(\sec x) = \tan x$.

259. A student writes $\dfrac{d}{dx}e^{x^{2}} = x^{2}e^{x^{2}-1}$. Explain the error and give the correct derivative.

260. Find the exact gradient of the tangent to $y = x\ln x - x$ at the point where $x = e$.

261. Let $f(x) = \dfrac{\ln x}{x^2}$, for $x > 0$.

(a) Show that $f'(x) = \dfrac{1 - 2\ln x}{x^3}$. (1 mark)

(b) Find the exact coordinates of the point where the tangent to $y = f(x)$ is horizontal. (2 marks)

Revision

262. The points $A(-1, 1)$ and $B(2, 4)$ lie on the parabola $y = x^2$.

(a) Find the equation of the tangent to the parabola at A. (2 marks)

(b) There is a point C on the parabola where the tangent is parallel to AB. Find the coordinates of C. (2 marks)

(c) Show that the tangents at A and B meet at a point with the same x-coordinate as C. (1 mark)

263. Let $f(x) = x - \dfrac{x^2}{2} + \dfrac{x^3}{3}$ and $g(x) = \ln(1 + x)$, for $x \geq 0$.

(a) Show that $1 - x + x^2 - \dfrac{1}{1+x} = \dfrac{x^3}{1+x}$. (1 mark)

(b) Use part (a) to show that $f'(x) \geq g'(x)$ for all $x \geq 0$. (2 marks)

(c) Find the equation of the tangent to $y = g(x)$ at the origin. (2 marks)

264. Let $f(x) = \tan x - x$, for $-\dfrac{\pi}{2} < x < \dfrac{\pi}{2}$.

(a) Show that $f'(x) = \tan^2 x$. (2 marks)

(b) Explain why $f(x)$ is never decreasing. (1 mark)

(c) Find the equation of the tangent to $y = f(x)$ at the point where $x = \dfrac{\pi}{4}$. (2 marks)

Solutions

1. a) $f(x) = 9x$, $f'(x) = 9$ b) $f(x) = 2 - x$, $f'(x) = -1$ c) $f(x) = x^4 - 3x$, $f'(x) = 4x^3 - 3$

2. a) $7e^{7x}$ b) $2x\,e^{x^2+1}$ c) $(3x^2 - 6)\,e^{x^3-6x} = 3(x^2 - 2)\,e^{x^3-6x}$

3. a) $-5e^{1-5x}$ b) $\dfrac{1}{2\sqrt{x}}\,e^{\sqrt{x}} = \dfrac{e^{\sqrt{x}}}{2\sqrt{x}}$ c) 0, since e^4 is a constant

4. a) $12e^{3x}$ b) $-10e^{5x+1}$ c) $3e^{6x}$

5. a) $6x\,e^{x^2}$ b) $-(4x - 1)\,e^{2x^2-x} = (1 - 4x)\,e^{2x^2-x}$

6. $\dfrac{d}{dx}\left(\frac{1}{4}e^{8x-2}\right) = \frac{1}{4} \times 8e^{8x-2} = 2e^{8x-2}$

7. a) $4e^{4x} - 4e^{-4x}$ b) $6e^{2x} - 3x^2$

8. $\dfrac{dy}{dx} = -6e^{-3x} - 5e^x + 7$

9. $\dfrac{dy}{dx} = 2x - 4 - e^{2-x}$

10. a) $y = e^{-x}$, so $\dfrac{dy}{dx} = -e^{-x}$ b) $y = 4e^{-2x}$, so $\dfrac{dy}{dx} = -8e^{-2x}$

11. a) $y = e^{4x}$, so $\dfrac{dy}{dx} = 4e^{4x}$ b) $y = e^{6x}$, so $\dfrac{dy}{dx} = 6e^{6x}$

12. a) $y = e^{2x}$, so $\dfrac{dy}{dx} = 2e^{2x}$ b) $y = e^{2x} + 2 + e^{-2x}$, so $\dfrac{dy}{dx} = 2e^{2x} - 2e^{-2x}$

13. $f'(x) = 2e^{2x+1}$, so $f'(0) = 2e$.

14. $f'(x) = 2x\,e^{x^2-4}$, so $f'(2) = 4e^0 = 4$.

15. $\dfrac{dy}{dx} = -3e^{-x} + e^x$. At $x = \ln 3$: $e^{\ln 3} = 3$ and $e^{-\ln 3} = \frac{1}{3}$, so the gradient is $-3 \times \frac{1}{3} + 3 = 2$.

16. $\dfrac{dy}{dx} = 3\left(e^x + 1\right)^2 \times e^x = 3e^x\left(e^x + 1\right)^2$

17. $\dfrac{dy}{dx} = 4\left(2 - e^{3x}\right)^3 \times \left(-3e^{3x}\right) = -12e^{3x}\left(2 - e^{3x}\right)^3$

18. $y = \left(1 + e^{2x}\right)^{\frac{1}{2}}$, so $\dfrac{dy}{dx} = \frac{1}{2}\left(1 + e^{2x}\right)^{-\frac{1}{2}} \times 2e^{2x} = \dfrac{e^{2x}}{\sqrt{1 + e^{2x}}}$

19. a) $y = e^{x^2-2x}$, so $\dfrac{dy}{dx} = (2x - 2)\,e^{x^2-2x} = 2(x - 1)\,e^{x^2-2x}$. b) $e^{x^2-2x} > 0$, so $x = 1$; $y = e^{-1}$. The point is $\left(1,\, e^{-1}\right)$.

20. a) $y = e^{2x^2-8x}$, so $\dfrac{dy}{dx} = (4x - 8)\,e^{2x^2-8x} = 4(x - 2)\,e^{2x^2-8x}$. b) $e^{2x^2-8x} > 0$, so $x = 2$; $y = e^{8-16} = e^{-8}$. The point is $\left(2,\, e^{-8}\right)$.

21. a) $f'(x) = (3x^2-3)\,e^{x^3-3x} = 3(x-1)(x+1)\,e^{x^3-3x}$ b) $x = 1$ or $x = -1$. $f(1) = e^{-2}$ and $f(-1) = e^2$, giving $(1,\, e^{-2})$ and $(-1,\, e^2)$.

22. The student has used the power rule, which only applies when x is in the base. Here x is in the power, so the power stays the same and is multiplied by its derivative: $\dfrac{d}{dx}\left(e^{3x}\right) = 3e^{3x}$.

23. Let $u = ax + b$, so $y = e^u$. Then $\dfrac{du}{dx} = a$ and $\dfrac{dy}{du} = e^u$, so $\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx} = e^u \times a = ae^{ax+b}$.

24. $\dfrac{dy}{dx} = 10e^{2x}$. Since $e^{2x} > 0$ for all x, the gradient is always positive, so the graph is always increasing.

25. $\dfrac{dy}{dx} = (2x+k)\,e^{x^2+kx}$. The exponential is never 0, so $2(3) + k = 0$ and $k = -6$.

26. a) $\dfrac{dy}{dx} = 10e^{2x}$ b) $10e^{2x} = 2 \times 5e^{2x} = 2y$ c) $\dfrac{dy}{dx} = 2 \times 10 = 20$

27. (a) $f'(x) = (2x-1)^4 + x \times 4(2x-1)^3 \times 2 = (2x-1)^3(2x-1+8x) = (2x-1)^3(10x-1)$ (b) $f(1) = 1$, $f'(1) = 1 \times 9 = 9$: $y - 1 = 9(x-1)$, so $y = 9x - 8$. (c) $f'(x) = 0$: $x = \frac{1}{2}$ or $x = \frac{1}{10}$. *cf. 2008 HSC Q2(a)*

28. (a) Let $u = \ln x$: $u^2 - u - 2 = 0$, so $(u-2)(u+1) = 0$ and $x = e^2$ or $x = e^{-1}$. (b) $2(0.36) + 0.57 = 1.29$ (c) $\log_a \frac{3}{2} = 0.57 - 0.36 = 0.21$ *cf. 2008 HSC Q7(a)*

29. (a) First game X or G; second game X or G; a third game only after XG or GX. Outcomes: XX, XGX, XGG, GXX, GXG, GG. (b) $GG + XGG + GXG = \dfrac{4}{25} + 2 \times \dfrac{3}{5} \times \dfrac{2}{5} \times \dfrac{2}{5} = \dfrac{20}{125} + \dfrac{24}{125} = \dfrac{44}{125}$ (c) The first two games are split: $2 \times \dfrac{3}{5} \times \dfrac{2}{5} = \dfrac{12}{25}$ *cf. 2008 HSC Q7(c)*

30. a) $e^{x \ln 9}$ b) $e^{5x \ln 2}$

31. a) $x^2e^{2x}(3+2x)$ b) $e^{3x}(1-3x)$

32. a) $(\ln 6)\,6^x$ b) $5(\ln 2)\,2^x$ c) $3^{4x} = e^{4x \ln 3}$, so $\dfrac{dy}{dx} = 4(\ln 3)\,3^{4x}$

33. a) $e^x + xe^x = e^x(1+x)$ b) $2xe^{4x} + 4x^2e^{4x} = 2xe^{4x}(1+2x)$

34. $\dfrac{dy}{dx} = 4x^3e^x + x^4e^x = x^3e^x(4+x)$

35. $\dfrac{dy}{dx} = 3e^{-x} - (3x-1)e^{-x} = e^{-x}(4-3x)$

36. $\dfrac{dy}{dx} = e^{-2x} - 2xe^{-2x} = e^{-2x}(1-2x)$

37. $\dfrac{dy}{dx} = 2xe^{-3x} - 3x^2e^{-3x} = xe^{-3x}(2-3x)$

38. $\dfrac{dy}{dx} = \dfrac{3e^{3x} \cdot x - e^{3x} \cdot 1}{x^2} = \dfrac{e^{3x}(3x-1)}{x^2}$

39. $\dfrac{dy}{dx} = \dfrac{1 \cdot e^{2x} - x \cdot 2e^{2x}}{(e^{2x})^2} = \dfrac{e^{2x}(1-2x)}{e^{4x}} = \dfrac{1-2x}{e^{2x}}$

40. $\dfrac{dy}{dx} = \dfrac{e^x(x+1) - e^x \cdot 1}{(x+1)^2} = \dfrac{xe^x}{(x+1)^2}$

41. $\dfrac{dy}{dx} = 2^x + x(\ln 2)\,2^x = 2^x(1 + x\ln 2)$

42. $\dfrac{dy}{dx} = 2x \cdot 5^x + x^2(\ln 5)\,5^x = x \cdot 5^x(2 + x\ln 5)$

43. $\dfrac{dy}{dx} = (\ln 2)\,2^x e^x + 2^x e^x = 2^x e^x(\ln 2 + 1)$

44. $\dfrac{dy}{dx} = \dfrac{e^x(e^x+2) - e^x \cdot e^x}{(e^x+2)^2} = \dfrac{2e^x}{(e^x+2)^2}$

45. $\dfrac{dy}{dx} = \dfrac{e^x(e^x+1) - (e^x-1)e^x}{(e^x+1)^2} = \dfrac{2e^x}{(e^x+1)^2}$

46. $\dfrac{dy}{dx} = \dfrac{2e^{2x}(1+e^{2x}) - e^{2x} \cdot 2e^{2x}}{(1+e^{2x})^2} = \dfrac{2e^{2x}}{(1+e^{2x})^2}$

47. $\dfrac{dy}{dx} = e^{-x}(1-x)$. $e^{-x} > 0$, so $x = 1$ and $y = e^{-1}$. The point is $\left(1,\, e^{-1}\right)$.

48. $\dfrac{dy}{dx} = xe^x(x+2)$. $e^x > 0$, so $x = 0$ or $x = -2$, giving $(0,\, 0)$ and $\left(-2,\, 4e^{-2}\right)$.

49. $\dfrac{dy}{dx} = \dfrac{e^x(x-1)}{x^2}$, which is 0 when $x = 1$. $y = e$. The point is $(1,\, e)$.

50. $a = e^{\ln a}$, so $a^x = e^{x\ln a}$. Since $\ln a$ is a constant, $\dfrac{d}{dx}\left(e^{x\ln a}\right) = \ln a \cdot e^{x\ln a} = (\ln a)\,a^x$.

51. The student used the power rule, which only applies when x is the base. Here x is the power, so $\dfrac{d}{dx}\left(2^x\right) = (\ln 2)\,2^x$.

52. $\dfrac{d}{dx}\left(xe^x\right) = e^x + xe^x = (x+1)e^x$. At $x = 0$: $(0+1)e^0 = 1$.

53. $\dfrac{dy}{dx} = e^{2x}(1+2x)$. At $x = 1$: $e^2(1+2) = 3e^2$.

54. a) $f'(x) = e^{-x} - xe^{-x} = (1-x)e^{-x}$ b) For $x < 1$, $1 - x > 0$ and $e^{-x} > 0$, so $f'(x) > 0$. c) $f'(x) = 0$ when $x = 1$; $f(1) = e^{-1}$. The point is $\left(1,\, e^{-1}\right)$.

55. (a) $f'(x) = (2x-2)e^{x^2-2x}$ (b) $e^{x^2-2x} > 0$, so $2x - 2 = 0$: $x = 1$, $y = e^{-1}$. The point is $\left(1,\, \dfrac{1}{e}\right)$. (c) $f(2) = e^0 = 1$, $f'(2) = 2$: $y - 1 = 2(x-2)$, so $y = 2x - 3$.

56. (a) $\Delta = (k+2)^2 - 4(k+5) = k^2 - 16 = 0$, so $k = \pm 4$. (b) $x^2 - 6x + 9 = 0$, so $(x-3)^2 = 0$ and $x = 3$. (c) $k^2 - 16 < 0$: $-4 < k < 4$. *cf. 2009 HSC Q4(b)*

57. (a) $\frac{1}{2} \times 2 \times 2\sin\theta = \sqrt{3}$, so $\sin\theta = \dfrac{\sqrt{3}}{2}$; the other value is $\dfrac{2\pi}{3}$. (b) $\frac{1}{2} \times 4 \times \dfrac{\pi}{3} = \dfrac{2\pi}{3}$ cm^2 (c) Triangle AOB is equilateral, so $AB = 2$; arc $= 2 \times \dfrac{\pi}{3}$. Perimeter $= 2 + \dfrac{2\pi}{3}$ cm. *cf. 2009 HSC Q5(c)*

58. a) e^4 b) $3e^0 = 3$ c) e^3

59. a) $-\dfrac{1}{5e}$ b) $-\dfrac{1}{e^4}$ c) $\dfrac{e}{3}$

60. Point $(2, e^2)$, $m = e^2$. $y - e^2 = e^2(x-2)$, so $y = e^2x - e^2$.

61. Point $(0, 1)$, $m = 5e^0 = 5$. $y - 1 = 5x$, so $y = 5x + 1$.

62. Point $(1, 2e^3)$, $m = 6e^3$. $y - 2e^3 = 6e^3(x-1)$, so $y = 6e^3x - 4e^3$.

63. Point $(0, 1)$. Tangent gradient 3, normal gradient $-\frac{1}{3}$. $y - 1 = -\frac{1}{3}x$, so $x + 3y - 3 = 0$.

64. Point $(1, e)$. Tangent gradient e, normal gradient $-\dfrac{1}{e}$. $e(y-e) = -(x-1)$, so $x + ey - e^2 - 1 = 0$.

65. Point $(0, 1)$. Tangent gradient -2, normal gradient $\frac{1}{2}$. $y - 1 = \frac{1}{2}x$, so $x - 2y + 2 = 0$.

66. $\dfrac{dy}{dx} = e^{2x}(1+2x)$, so $m = 1$ at $x = 0$. The tangent is $y = x$.

67. Point $(2, 2e^{-2})$. $\dfrac{dy}{dx} = e^{-x}(1-x)$, so $m = -e^{-2}$. $y - 2e^{-2} = -e^{-2}(x-2)$, so $y = -e^{-2}x + 4e^{-2}$.

68. Point $(1, e)$. $\dfrac{dy}{dx} = xe^x(2+x)$, so $m = 3e$. $y - e = 3e(x-1)$, so $y = 3ex - 2e$.

69. Point $(-1, e)$. $\dfrac{dy}{dx} = 2xe^{x^2} = -2e$, so the normal gradient is $\dfrac{1}{2e}$. $2e(y-e) = x + 1$, so $x - 2ey + 2e^2 + 1 = 0$.

70. Point $(1, 1)$. Tangent gradient $2e^0 = 2$, normal gradient $-\frac{1}{2}$. $y - 1 = -\frac{1}{2}(x-1)$, so $x + 2y - 3 = 0$.

71. Point $(2, 1)$. Tangent gradient $2(2)e^0 = 4$, normal gradient $-\frac{1}{4}$. $y - 1 = -\frac{1}{4}(x-2)$, so $x + 4y - 6 = 0$.

72. Point (a, e^{2a}); $\dfrac{dy}{dx} = 2e^{2x}$, so $m = 2e^{2a}$. Tangent: $y - e^{2a} = 2e^{2a}(x-a)$. At $y = 0$: $-e^{2a} = 2e^{2a}(x-a)$. Since $e^{2a} \neq 0$, $x - a = -\frac{1}{2}$, so $x = a - \frac{1}{2}$.

73. Point (a, e^{-a}); $\dfrac{dy}{dx} = -e^{-x}$, so $m = -e^{-a}$. Tangent: $y - e^{-a} = -e^{-a}(x-a)$. At $y = 0$: $-e^{-a} = -e^{-a}(x-a)$. Since $e^{-a} \neq 0$, $x - a = 1$, so $x = a + 1$.

74. Tangent: $y - e^a = e^a(x-a)$. At $x = 0$: $y = e^a - ae^a = (1-a)e^a$.

75. Tangent: $y - e^{2a} = 2e^{2a}(x-a)$. Through $(0, 0)$: $-e^{2a} = -2ae^{2a}$, and $e^{2a} > 0$, so $a = \frac{1}{2}$. The point is $\left(\frac{1}{2}, e\right)$ with $m = 2e$, so the tangent is $y = 2ex$.

76. $e^x = 3$, so $x = \ln 3$ and $y = 3$. The point is $(\ln 3, 3)$.

77. The tangent gradient is 6: $3e^{3x} = 6$, so $e^{3x} = 2$ and $x = \frac{1}{3}\ln 2$. Then $y = 2$. The point is $\left(\frac{1}{3}\ln 2, 2\right)$.

78. A horizontal tangent has gradient 0. The gradient is $\dfrac{dy}{dx} = e^x$, which is greater than 0 for every x, so it is never 0.

79. For $y = e^x$: $m_1 = e^0 = 1$. For $y = e^{-x}$: $m_2 = -e^0 = -1$. $m_1 m_2 = -1$, so the tangents are perpendicular.

80. Point $(0, 1)$; tangent gradient 1, normal gradient -1. Normal: $y = -x + 1$, which crosses at $(1, 0)$.

81. The y-value must come from the original function, $y = e^2$, not from the derivative $2e^2$. The correct point is $\left(1, e^2\right)$.

82. a) Point $(0, 1)$, $f'(x) = 2e^{2x}$ so $m = 2$. Tangent: $y = 2x + 1$. b) $2x + 1 = 0$, so $x = -\frac{1}{2}$: the point $\left(-\frac{1}{2}, 0\right)$.

83. (a) Product rule: $e^{-2x} + x \times (-2e^{-2x}) = e^{-2x}(1 - 2x)$ (b) $e^{-2x} > 0$, so $x = \frac{1}{2}$ and $y = \frac{1}{2}e^{-1}$: $\left(\frac{1}{2}, \frac{1}{2e}\right)$. (c) $f'(x) > 0$ when $1 - 2x > 0$: $x < \frac{1}{2}$. *cf. 2011 HSC Q2(d)*

84. (a) $A(0, -3)$, $B(1, 0)$: $AB = \sqrt{1 + 9} = \sqrt{10}$ (b) Gradient $-\frac{1}{3}$ through B: $y = -\frac{1}{3}(x - 1)$, i.e. $x + 3y - 1 = 0$. (c) $C\left(0, \frac{1}{3}\right)$, so $AC = \frac{10}{3}$ and the height from B is 1: area $= \frac{1}{2} \times \frac{10}{3} \times 1 = \frac{5}{3}$ square units. *cf. 2009 HSC Q5(a)*

85. (a) $2x + 1 = 4$, so $x = \frac{3}{2}$. (b) Let $u = 2^x$: $(u - 4)(u + 1) = 0$. $2^x = 4$ gives $x = 2$; $2^x = -1$ has no solution since $2^x > 0$. (c) $x = \dfrac{\ln 10}{\ln 2} = 3.32$ *cf. 2011 HSC Q1(c), 2008 HSC Q7(a)*

86. a) $f(x) = 9x$, $f'(x) = 9$ b) $f(x) = x^2 - 7$, $f'(x) = 2x$ c) $f(x) = 5 - 2x$, $f'(x) = -2$

87. a) $3\ln x + \ln(x + 2)$ b) $\ln(x - 1) - \ln(x + 5)$ c) $2\ln(4x + 3)$

88. a) Yes (product) b) No (a sum) c) Yes (power)

89. a) $\dfrac{2x}{x^2 + 1}$ b) $\dfrac{3x^2 - 4}{x^3 - 4x}$

90. $\dfrac{dy}{dx} = \dfrac{4x - 5}{2x^2 - 5x + 3}$

91. $\dfrac{dy}{dx} = \dfrac{6}{6x} = \dfrac{1}{x}$

92. $\dfrac{dy}{dx} = \dfrac{3}{3x + 2}$

93. $\dfrac{dy}{dx} = \dfrac{-4}{1 - 4x} = \dfrac{4}{4x - 1}$

94. $\dfrac{dy}{dx} = 5 \times \dfrac{2}{2x + 3} = \dfrac{10}{2x + 3}$

95. $\dfrac{dy}{dx} = 3x^2 - \dfrac{2}{x}$

96. $\dfrac{dy}{dx} = \dfrac{2x}{x^2 + 1} + 2e^{2x}$

97. $y = 3\ln(2x + 1)$, so $\dfrac{dy}{dx} = \dfrac{6}{2x + 1}$.

98. $y = \frac{1}{2}\ln(4x-1)$, so $\dfrac{dy}{dx} = \frac{1}{2} \times \dfrac{4}{4x-1} = \dfrac{2}{4x-1}$.

99. $y = 5\ln x$, so $\dfrac{dy}{dx} = \dfrac{5}{x}$.

100. $y = \ln x + \ln(x+3)$, so $\dfrac{dy}{dx} = \dfrac{1}{x} + \dfrac{1}{x+3} = \dfrac{2x+3}{x(x+3)}$.

101. $\dfrac{dy}{dx} = \dfrac{1}{x+2} - \dfrac{1}{x-1} = \dfrac{(x-1)-(x+2)}{(x+2)(x-1)} = \dfrac{-3}{(x+2)(x-1)}$

102. $\dfrac{dy}{dx} = \dfrac{3}{3x-1} - \dfrac{1}{x+1} = \dfrac{3(x+1)-(3x-1)}{(3x-1)(x+1)} = \dfrac{4}{(3x-1)(x+1)}$

103. $y = 3\ln x + \frac{1}{2}\ln(x+1)$, so $\dfrac{dy}{dx} = \dfrac{3}{x} + \dfrac{1}{2(x+1)} = \dfrac{7x+6}{2x(x+1)}$.

104. $y = 2\ln(x+1) - \ln(x-2)$, so $\dfrac{dy}{dx} = \dfrac{2}{x+1} - \dfrac{1}{x-2} = \dfrac{x-5}{(x+1)(x-2)}$.

105. $y = \frac{1}{2}\ln(x+1) - \frac{1}{2}\ln(x-1)$, so $\dfrac{dy}{dx} = \frac{1}{2}\left(\dfrac{1}{x+1} - \dfrac{1}{x-1}\right) = \frac{1}{2} \times \dfrac{-2}{x^2-1} = -\dfrac{1}{x^2-1}$.

106. Differentiate both sides with the chain rule: $e^{\ln x} \times \dfrac{d}{dx}(\ln x) = 1$. Since $e^{\ln x} = x$, this gives $x\dfrac{d}{dx}(\ln x) = 1$, so $\dfrac{d}{dx}(\ln x) = \dfrac{1}{x}$.

107. $\dfrac{d}{dx}\ln(5x) = \dfrac{5}{5x} = \dfrac{1}{x}$. By the log laws $\ln(5x) = \ln 5 + \ln x$, and $\ln 5$ is a constant with derivative 0.

108. The student forgot to multiply by the derivative of the inside. Correct: $\dfrac{2x}{x^2+1}$.

109. $\dfrac{dy}{dx} = \dfrac{2x}{x^2+3} = \dfrac{2}{4} = \dfrac{1}{2}$ at $x = 1$.

110. a) $f'(x) = \dfrac{2x-2}{x^2-2x+5}$ b) $x^2-2x+5 = (x-1)^2+4 > 0$ for all x, so the log is always defined. c) $f'(x) = 0$ when $x = 1$; $f(1) = \ln 4$. The point is $(1, \ln 4)$.

111. (a) Point $(1, e)$, $m = e$: $y - e = e(x-1)$, so $y = ex$. (b) $x = 0$ gives $y = 0$. (c) At $(0, 1)$ the tangent gradient is 1, so the normal is $y = -x + 1$, meeting the x-axis at $(1, 0)$.

112. (a) $f'(x) = 3x^2 + 6x + k$ (b) Need $f'(x) \geq 0$ for all x: $\Delta = 36 - 12k \leq 0$, so $k \geq 3$. (c) $3x^2 + 6x - 9 = 0$, so $x^2 + 2x - 3 = 0$ and $x = 1$ or $x = -3$. *cf. 2010 HSC Q8(d)*

113. (a) $\dfrac{1}{4}$ (b) $\dfrac{3}{4} \times \dfrac{2}{3} \times \dfrac{1}{2} = \dfrac{1}{4}$ (c) $\left(\dfrac{3}{4}\right)^5 = \dfrac{243}{1024}$ (d) $1 - \dfrac{243}{1024} = \dfrac{781}{1024}$ *cf. 2009 HSC Q5(b)*

114. a) $\dfrac{\ln x}{\ln 6}$ b) $\dfrac{\ln(x^2-1)}{\ln 3}$

115. a) $\dfrac{1}{x\ln 4}$ b) $\dfrac{3}{x\ln 5}$ c) $\dfrac{3}{(3x-1)\ln 2}$

116. a) Log law b) Chain rule c) Chain rule d) Log law

117. $\dfrac{dy}{dx} = 1 \times \ln x + x \times \dfrac{1}{x} = \ln x + 1$

118. $\dfrac{dy}{dx} = 5x^4 \ln x + x^4 = x^4(5\ln x + 1)$

119. $\dfrac{dy}{dx} = e^x \ln x + \dfrac{e^x}{x} = e^x\left(\ln x + \dfrac{1}{x}\right)$

120. $\dfrac{dy}{dx} = 2x\ln(3x) + x^2 \times \dfrac{1}{x} = x\big(2\ln(3x) + 1\big)$

121. $\dfrac{dy}{dx} = \dfrac{\ln x}{2\sqrt{x}} + \dfrac{\sqrt{x}}{x} = \dfrac{\ln x}{2\sqrt{x}} + \dfrac{2}{2\sqrt{x}} = \dfrac{\ln x + 2}{2\sqrt{x}}$

122. $\dfrac{dy}{dx} = \dfrac{\frac{1}{x}\cdot x^2 - \ln x \cdot 2x}{x^4} = \dfrac{x(1 - 2\ln x)}{x^4} = \dfrac{1 - 2\ln x}{x^3}$

123. $\dfrac{dy}{dx} = \dfrac{1\cdot \ln x - x\cdot \frac{1}{x}}{(\ln x)^2} = \dfrac{\ln x - 1}{(\ln x)^2}$

124. $\dfrac{dy}{dx} = \dfrac{\frac{1}{x}(x+1) - \ln x}{(x+1)^2} = \dfrac{x + 1 - x\ln x}{x(x+1)^2}$

125. $\dfrac{dy}{dx} = 2\ln x \times \dfrac{1}{x} = \dfrac{2\ln x}{x}$

126. $\dfrac{dy}{dx} = 4(\ln x)^3 \times \dfrac{1}{x} = \dfrac{4(\ln x)^3}{x}$

127. $y = (\ln x)^{\frac{1}{2}}$, so $\dfrac{dy}{dx} = \frac{1}{2}(\ln x)^{-\frac{1}{2}} \times \dfrac{1}{x} = \dfrac{1}{2x\sqrt{\ln x}}$.

128. Point $(1, 0)$, $m = 1$. The tangent is $y = x - 1$.

129. Point $\left(\frac{1}{3}, \ln 1\right) = \left(\frac{1}{3}, 0\right)$. $\dfrac{dy}{dx} = \dfrac{1}{x}$, so $m = 3$. $y = 3\left(x - \frac{1}{3}\right) = 3x - 1$.

130. Point $(e, 2)$. $\dfrac{dy}{dx} = \dfrac{2}{x}$, so $m = \dfrac{2}{e}$. $y - 2 = \dfrac{2}{e}(x - e)$, so $y = \dfrac{2x}{e}$.

131. Point $(e, 1)$, tangent gradient $\dfrac{1}{e}$, normal gradient $-e$. $y - 1 = -e(x - e)$, so $y = -ex + e^2 + 1$.

132. Point $(1, 0)$. $\dfrac{dy}{dx} = x(2\ln x + 1)$, so $m = 1$. The tangent is $y = x - 1$.

133. Point $(1, 0)$. $\dfrac{dy}{dx} = \ln x + 1 = 1$, so the normal gradient is -1. The normal is $y = -x + 1$.

134. By the change of base law, $\log_a x = \dfrac{\ln x}{\ln a} = \dfrac{1}{\ln a}\cdot \ln x$. Since $\dfrac{1}{\ln a}$ is a constant, $\dfrac{d}{dx}(\log_a x) = \dfrac{1}{\ln a}\cdot\dfrac{1}{x} = \dfrac{1}{x\ln a}$.

135. $\ln\left(x^2\right) = 2\ln x$, with derivative $\dfrac{2}{x}$. $(\ln x)^2$ needs the chain rule: $\dfrac{2\ln x}{x}$. In the first the power is inside the log; in the second it is on the whole log.

136. Point $(e,\,1)$, $m = \dfrac{1}{e}$. $y - 1 = \dfrac{1}{e}(x - e)$ gives $y = \dfrac{x}{e}$, which passes through $(0,\,0)$.

137. $\dfrac{dy}{dx} = \ln x + 1 = 0$, so $\ln x = -1$ and $x = e^{-1}$. $y = e^{-1}\ln e^{-1} = -e^{-1}$. The point is $\left(e^{-1},\,-e^{-1}\right)$.

138. a) $f'(x) = \dfrac{\frac{1}{x}\cdot x - \ln x \cdot 1}{x^2} = \dfrac{1 - \ln x}{x^2}$ b) $1 - \ln x = 0$, so $x = e$; $f(e) = \dfrac{1}{e}$. The point is $\left(e,\,\dfrac{1}{e}\right)$.

139. (a) $f(x) = \ln(x+1) - \ln(x-1)$, so $f'(x) = \dfrac{1}{x+1} - \dfrac{1}{x-1} = \dfrac{(x-1)-(x+1)}{x^2-1} = -\dfrac{2}{x^2-1}$
(b) For $x > 1$, $x^2 - 1 > 0$, so $f'(x) < 0$. (c) $f(3) = \ln 2$, $f'(3) = -\frac{1}{4}$: $y - \ln 2 = -\frac{1}{4}(x-3)$, so $y = -\frac{1}{4}x + \frac{3}{4} + \ln 2$.

140. (a) $(e^x - 1)(e^x - 2) = 0$, so $x = 0$ or $x = \ln 2$. (b) Let $u = \log_2 x$: $u^2 - 4u + 3 = 0$, so $u = 1$ or 3, giving $x = 2$ or $x = 8$. *cf. 2008 HSC Q7(a)*

141. (a) $A = 3$; $\dfrac{2\pi}{b} = \pi$, so $b = 2$. (b) $\sin 2x = \dfrac{\sqrt{3}}{2}$ with $0 \le 2x \le 2\pi$: $2x = \dfrac{\pi}{3}$ or $\dfrac{2\pi}{3}$, so $x = \dfrac{\pi}{6}$ or $\dfrac{\pi}{3}$. *cf. 2010 HSC Q8(c)*

142. a) $f(x) = 9x$, $f'(x) = 9$ b) $f(x) = x^2 - 1$, $f'(x) = 2x$ c) $f(x) = 5 - 2x$, $f'(x) = -2$

143. a) $\cos x - \sin x$ b) $4\sec^2 x$ c) $2x + 3\sin x$

144. $\dfrac{dy}{dx} = (2x+3)\cos\left(x^2+3x\right)$

145. $\dfrac{dy}{dx} = 3x^2\cos\left(x^3\right)$

146. $\dfrac{dy}{dx} = 5\cos(5x-2)$

147. $\dfrac{dy}{dx} = -2x\sin\left(x^2+1\right)$

148. $\dfrac{dy}{dx} = -4\sin 4x$

149. $\dfrac{dy}{dx} = -(-3)\sin(2-3x) = 3\sin(2-3x)$

150. $\dfrac{dy}{dx} = 6\sec^2 6x$

151. $\dfrac{dy}{dx} = 2x\sec^2\left(x^2\right)$

152. $\dfrac{dy}{dx} = -\sec^2(1-x)$

153. $\dfrac{dy}{dx} = 15\cos 3x - 8\sin 4x$

154. $\dfrac{dy}{dx} = 2 \times \frac{1}{2}\sec^2\dfrac{x}{2} = \sec^2\dfrac{x}{2}$

155. $\dfrac{dy}{dx} = -\pi\sin(\pi x) - \pi\cos(\pi x)$

156. $\dfrac{dy}{dx} = -2\sin 2x$. At $x = \dfrac{\pi}{12}$: $-2\sin\dfrac{\pi}{6} = -1$.

157. $\dfrac{dy}{dx} = \sec^2 x = \dfrac{1}{\cos^2 x}$. At $x = \dfrac{\pi}{4}$: $\dfrac{1}{\left(\frac{1}{\sqrt{2}}\right)^2} = 2$.

158. $\dfrac{dy}{dx} = 3\cos 3x$. At $x = \dfrac{\pi}{9}$: $3\cos\dfrac{\pi}{3} = \dfrac{3}{2}$.

159. $\dfrac{dy}{dx} = \cos x + \sin x = 0$, so $\tan x = -1$: $x = \dfrac{3\pi}{4}$ or $\dfrac{7\pi}{4}$.

160. $\dfrac{dy}{dx} = 1 - 2\sin x = 0$, so $\sin x = \frac{1}{2}$: $x = \dfrac{\pi}{6}$ or $\dfrac{5\pi}{6}$.

161. $\dfrac{dy}{dx} = 2\cos 2x = 0$, so $2x = \dfrac{\pi}{2}$ or $\dfrac{3\pi}{2}$: $x = \dfrac{\pi}{4}$ or $\dfrac{3\pi}{4}$.

162. $\tan x = \dfrac{\sin x}{\cos x}$, so $\dfrac{d}{dx}(\tan x) = \dfrac{\cos x \cdot \cos x - \sin x \cdot (-\sin x)}{\cos^2 x} = \dfrac{\cos^2 x + \sin^2 x}{\cos^2 x} = \dfrac{1}{\cos^2 x} = \sec^2 x$.

163. At $x = 0$ the curve rises most steeply, with gradient 1; at $x = \frac{\pi}{2}$ it reaches a maximum, so the tangent is horizontal; at $x = \pi$ it falls most steeply, with gradient -1. These are the values of $\cos x$ at 0, $\frac{\pi}{2}$ and π.

164. The student forgot to multiply by the derivative of the angle, 3. Correct: $-3\sin 3x$.

165. $\dfrac{dy}{dx} = \sec^2 x = \dfrac{1}{\cos^2 x}$. Since $0 < \cos^2 x \leq 1$ wherever $\tan x$ is defined, the gradient is always at least 1, so it is never 0.

166. a) $f'(x) = \cos x - \sqrt{3}\sin x$ b) $\cos x = \sqrt{3}\sin x$, so $\tan x = \dfrac{1}{\sqrt{3}}$: $x = \dfrac{\pi}{6}$ or $\dfrac{7\pi}{6}$.

167. (a) $3x^2\ln x + x^3 \times \dfrac{1}{x} = x^2(3\ln x + 1)$ (b) $x^2 > 0$, so $\ln x = -\frac{1}{3}$ and $x = e^{-\frac{1}{3}}$. (c) $f(e) = e^3$, $f'(e) = 4e^2$: $y - e^3 = 4e^2(x - e)$, so $y = 4e^2x - 3e^3$. *cf. 2008 HSC Q2(a)(ii)*

168. (a) $\dfrac{1}{\cos^2 x} + \dfrac{\sin x}{\cos^2 x} = \dfrac{1 + \sin x}{\cos^2 x}$ (b) $\cos^2 x = (1 - \sin x)(1 + \sin x)$, so the expression is $\dfrac{1}{1 - \sin x}$.
(c) $1 - \sin x = \frac{1}{2}$, so $\sin x = \frac{1}{2}$: $x = \dfrac{\pi}{6}$ or $\dfrac{5\pi}{6}$. *cf. 2010 HSC Q5(b)*

169. (a) $\frac{1}{2} \times 9 \times \dfrac{\pi}{3} = \dfrac{3\pi}{2}$ cm^2 (b) $AB = 3$ (equilateral triangle), arc $= \pi$: perimeter $3 + \pi$ cm. (c) $\dfrac{3\pi}{2} - \frac{1}{2} \times 9\sin\dfrac{\pi}{3} = \dfrac{3\pi}{2} - \dfrac{9\sqrt{3}}{4}$ cm^2 *cf. 2009 HSC Q5(c)(ii)*

170. a) $(\cos x)^3$ b) $(\sin 4x)^2$

171. $\dfrac{dy}{dx} = \sin x + x\cos x$

172. $\dfrac{dy}{dx} = 2x\cos x - x^2\sin x = x(2\cos x - x\sin x)$

173. $\dfrac{dy}{dx} = 3x^2\tan x + x^3\sec^2 x = x^2\left(3\tan x + x\sec^2 x\right)$

174. $\dfrac{dy}{dx} = \dfrac{\sin x}{2\sqrt{x}} + \sqrt{x}\cos x$

175. $\dfrac{dy}{dx} = \sin 3x + 3x\cos 3x$

176. $\dfrac{dy}{dx} = 2x\cos 2x - 2x^2\sin 2x = 2x(\cos 2x - x\sin 2x)$

177. $\dfrac{dy}{dx} = \cos x\cdot\cos x + \sin x\cdot(-\sin x) = \cos^2 x - \sin^2 x$

178. $\dfrac{dy}{dx} = 2\cos x\times(-\sin x) = -2\sin x\cos x$

179. $\dfrac{dy}{dx} = 3\sin^2 x\cos x$

180. $\dfrac{dy}{dx} = 2\tan x\sec^2 x$

181. $\dfrac{dy}{dx} = \dfrac{-x\sin x - \cos x}{x^2}$

182. $\dfrac{dy}{dx} = \dfrac{\sin x - x\cos x}{\sin^2 x}$

183. $\dfrac{\cos x(1+\cos x) - \sin x(-\sin x)}{(1+\cos x)^2} = \dfrac{\cos x + \cos^2 x + \sin^2 x}{(1+\cos x)^2} = \dfrac{1+\cos x}{(1+\cos x)^2} = \dfrac{1}{1+\cos x}$

184. $\dfrac{dy}{dx} = e^x\sin x + e^x\cos x = e^x(\sin x + \cos x)$

185. $\dfrac{dy}{dx} = \dfrac{\cos x}{\sin x}$

186. $\dfrac{dy}{dx} = -\sin x\, e^{\cos x}$

187. Point $(\pi, -\pi)$. $\dfrac{dy}{dx} = \cos x - x\sin x = -1$. $y + \pi = -(x-\pi)$, so $y = -x$.

188. Point $\left(\dfrac{\pi}{4}, \dfrac{1}{2}\right)$. $\dfrac{dy}{dx} = 2\sin x\cos x = 2\times\dfrac{1}{\sqrt{2}}\times\dfrac{1}{\sqrt{2}} = 1$. $y = x - \dfrac{\pi}{4} + \dfrac{1}{2}$.

189. Point $(0, 1)$. $\dfrac{dy}{dx} = e^x(\cos x - \sin x) = 1$, so the normal gradient is -1: $y = -x + 1$.

190. Quotient rule: $u = \cos x$, $u' = -\sin x$; $v = 1 - \sin x$, $v' = -\cos x$.
$\dfrac{-\sin x(1-\sin x) + \cos^2 x}{(1-\sin x)^2} = \dfrac{-\sin x + \sin^2 x + \cos^2 x}{(1-\sin x)^2} = \dfrac{1-\sin x}{(1-\sin x)^2} = \dfrac{1}{1-\sin x}$

191. $\dfrac{d}{dx}(x\sin x) = \sin x + x\cos x$ (product rule) and $\dfrac{d}{dx}(\cos x) = -\sin x$. Sum: $\sin x + x\cos x - \sin x = x\cos x$.

192. Product rule: $u = e^{-x}$, $u' = -e^{-x}$; $v = \sin x + \cos x$, $v' = \cos x - \sin x$.
$-e^{-x}(\sin x + \cos x) + e^{-x}(\cos x - \sin x) = e^{-x}(-2\sin x) = -2e^{-x}\sin x$

193. $2\sin x\cos x + 2\cos x(-\sin x) = 0$. This makes sense because $\sin^2 x + \cos^2 x = 1$, a constant.

194. $\dfrac{d}{dx}\ln(\sin x) = \dfrac{\cos x}{\sin x} = \cot x$

195. The derivative of a product is not the product of the derivatives. By the product rule, $\dfrac{d}{dx}(x\sin x) = \sin x + x\cos x$.

196. $\dfrac{dy}{dx} = 2x\sin x + x^2\cos x$. At $x = \pi$: $0 + \pi^2(-1) = -\pi^2$.

197. a) $f'(x) = -e^{-x}\sin x + e^{-x}\cos x = e^{-x}(\cos x - \sin x)$ b) $e^{-x} > 0$, so $\cos x = \sin x$, $\tan x = 1$: $x = \dfrac{\pi}{4}$ or $\dfrac{5\pi}{4}$.

198. (a) $2\cos x - 1$ (b) $\cos x = \frac{1}{2}$: $x = \dfrac{\pi}{3}$ or $\dfrac{5\pi}{3}$. (c) Point $(\pi, -\pi)$, $m = 2\cos\pi - 1 = -3$: $y + \pi = -3(x - \pi)$, so $y = -3x + 2\pi$.

199. (a) Quotient rule: $\dfrac{\frac{1}{x}\times x - \ln x \times 1}{x^2} = \dfrac{1 - \ln x}{x^2}$ (b) $\ln x = 1$, so $x = e$ and $y = \dfrac{1}{e}$: $\left(e, \dfrac{1}{e}\right)$. (c) $f(1) = 0$, $f'(1) = 1$: $y = x - 1$. *cf. 2008 HSC Q2(a)(iii)*

200. (a) Divide $\sin^2 x + \cos^2 x = 1$ by $\cos^2 x$. (b) $\sec^2 x - 1 = 2\sec x - 1$, so $\sec x(\sec x - 2) = 0$. $\sec x \neq 0$, so $\sec x = 2$, $\cos x = \frac{1}{2}$: $x = \dfrac{\pi}{3}$ or $\dfrac{5\pi}{3}$. *cf. 2010 HSC Q5(b)*

201. a) $(\sin 5x)^{-1}$ b) $\left(\cos\left(x^3\right)\right)^{-1}$

202. a) $\dfrac{2}{\sqrt{3}}$ b) $\dfrac{2}{\sqrt{3}}$ c) 1 d) 1

203. $\dfrac{d}{dx}(\cos 3x)^{-1} = -(\cos 3x)^{-2} \times (-3\sin 3x) = \dfrac{3\sin 3x}{\cos^2 3x} = 3\tan 3x\sec 3x$

204. $y = (\cos x)^{-2}$, so $\dfrac{dy}{dx} = -2(\cos x)^{-3}(-\sin x) = \dfrac{2\sin x}{\cos^3 x} = 2\tan x\sec^2 x$.

205. $\dfrac{dy}{dx} = 5\tan x\sec x$

206. $\dfrac{dy}{dx} = -\cot x\operatorname{cosec} x + 2x$

207. $\dfrac{dy}{dx} = -2\cot x\operatorname{cosec} x$

208. $\dfrac{d}{dx}(\sin 2x)^{-1} = -(\sin 2x)^{-2} \times 2\cos 2x = -2 \cdot \dfrac{\cos 2x}{\sin 2x} \cdot \dfrac{1}{\sin 2x} = -2\cot 2x\operatorname{cosec} 2x$

209. $\dfrac{dy}{dx} = -\operatorname{cosec}^2 x - 1$

210. $\dfrac{dy}{dx} = -4\operatorname{cosec}^2 x$

211. $\dfrac{d}{dx}(\tan x)^{-1} = -(\tan x)^{-2}\sec^2 x = -\dfrac{\cos^2 x}{\sin^2 x}\cdot\dfrac{1}{\cos^2 x} = -\dfrac{1}{\sin^2 x} = -\operatorname{cosec}^2 x$

212. $\dfrac{dy}{dx} = \tan x \sec x - \cot x \operatorname{cosec} x$

213. $\dfrac{dy}{dx} = -2\operatorname{cosec}^2 x - 3\tan x \sec x$

214. $\dfrac{dy}{dx} = \sec^2 x - \operatorname{cosec}^2 x$

215. $\dfrac{dy}{dx} = 2x\tan\left(x^2\right)\sec\left(x^2\right)$

216. $\dfrac{dy}{dx} = -3\operatorname{cosec}^2 3x$

217. $\dfrac{dy}{dx} = -2\cot(2x-1)\operatorname{cosec}(2x-1)$

218. $\dfrac{dy}{dx} = \operatorname{cosec} x - x\cot x \operatorname{cosec} x = \operatorname{cosec} x(1 - x\cot x)$

219. $\dfrac{dy}{dx} = 2x\cot x - x^2\operatorname{cosec}^2 x = x\left(2\cot x - x\operatorname{cosec}^2 x\right)$

220. $\dfrac{dy}{dx} = e^x\sec x + e^x\tan x\sec x = e^x\sec x(1+\tan x)$

221. Point $\left(\dfrac{\pi}{3}, 2\right)$. $m = \tan\dfrac{\pi}{3}\sec\dfrac{\pi}{3} = \sqrt{3}\times 2 = 2\sqrt{3}$. $y - 2 = 2\sqrt{3}\left(x - \dfrac{\pi}{3}\right)$, so $y = 2\sqrt{3}x - \dfrac{2\sqrt{3}\pi}{3} + 2$.

222. Tangent gradient $-\cot\dfrac{\pi}{4}\operatorname{cosec}\dfrac{\pi}{4} = -\sqrt{2}$, so the normal gradient is $\dfrac{1}{\sqrt{2}}$.

223. Point $\left(\dfrac{\pi}{2}, 0\right)$. $m = -\operatorname{cosec}^2\dfrac{\pi}{2} = -1$. The tangent is $y = -x + \dfrac{\pi}{2}$.

224. $\sec x = (\cos x)^{-1}$, so $\dfrac{d}{dx}(\sec x) = -(\cos x)^{-2}(-\sin x) = \dfrac{\sin x}{\cos x}\cdot\dfrac{1}{\cos x} = \tan x \sec x$.

225. $\cot x = \dfrac{\cos x}{\sin x}$, so $\dfrac{d}{dx}(\cot x) = \dfrac{-\sin^2 x - \cos^2 x}{\sin^2 x} = -\dfrac{1}{\sin^2 x} = -\operatorname{cosec}^2 x$.

226. $\dfrac{dy}{dx} = -\operatorname{cosec}^2 x = -\dfrac{1}{\sin^2 x}$. Since $\sin^2 x > 0$ wherever $\cot x$ is defined, the gradient is always negative.

227. The derivative of a reciprocal is not the reciprocal of the derivative. Using the chain rule on $(\cos x)^{-1}$ gives $\dfrac{\sin x}{\cos^2 x} = \tan x \sec x$.

228. a) $f'(x) = -(\cos x)^{-2}(-\sin x) = \dfrac{\sin x}{\cos^2 x}$ b) $\sin x = 0$ gives $x = 0$; the point is $(0, 1)$. c) For $0 < x < \dfrac{\pi}{2}$, $\sin x > 0$ and $\cos^2 x > 0$.

229. (a) $\dfrac{d}{dx}\ln f(x) = \dfrac{f'(x)}{f(x)} = \dfrac{\cos x}{\sin x}$ (b) $y = \ln\dfrac{1}{\sqrt{2}} = -\frac{1}{2}\ln 2$; $m = \dfrac{\cos\frac{\pi}{4}}{\sin\frac{\pi}{4}} = 1$. $y + \frac{1}{2}\ln 2 = x - \dfrac{\pi}{4}$, so $y = x - \dfrac{\pi}{4} - \frac{1}{2}\ln 2$. *cf. 2008 HSC Q3(b)(i)*

230. (a) $f(x) = x^3 + 2x^2 + 4x + 8$, so $f'(x) = 3x^2 + 4x + 4$, with $\Delta = 16 - 48 < 0$; $f'(x)$ is never 0. (b) $(-2, 0)$ and $(0, 8)$ (c) $f'(0) = 4$: $y = 4x + 8$. *cf. 2010 HSC Q6(a)*

231. (a) $x^2 - 2x + 3 - c = 0$ has equal roots: $4 - 4(3 - c) = 0$, so $c = 2$. (b) $x = 1$, $y = 4$: $(1, 4)$. (c) Normal: $y - 4 = -\frac{1}{2}(x - 1)$, i.e. $y = -\frac{1}{2}x + \frac{9}{2}$. With $y = x^2 + 3$: $2x^2 + x - 3 = 0$, so $(2x + 3)(x - 1) = 0$ and $Q = \left(-\frac{3}{2}, \frac{21}{4}\right)$. *cf. 2009 HSC Q4(b)*

232. a) e^x b) $\dfrac{1}{x}$ c) $\cos x$ d) $-\operatorname{cosec}^2 x$ e) $(\ln 3)\,3^x$

233. a) Product b) Chain c) Quotient d) Term by term

234. $\dfrac{dy}{dx} = e^x\cos 2x - 2e^x\sin 2x = e^x(\cos 2x - 2\sin 2x)$

235. $\dfrac{dy}{dx} = 2xe^{3x} + 3x^2e^{3x} = xe^{3x}(2 + 3x)$

236. $\dfrac{dy}{dx} = 2xe^{-x} - x^2e^{-x} = xe^{-x}(2 - x)$

237. $\dfrac{dy}{dx} = e^{-3x} - 3xe^{-3x} = e^{-3x}(1 - 3x)$

238. $\dfrac{dy}{dx} = -e^{-x}\cos x - e^{-x}\sin x = -e^{-x}(\cos x + \sin x)$

239. $\dfrac{dy}{dx} = \dfrac{\frac{1}{x}e^x - e^x\ln x}{e^{2x}} = \dfrac{\frac{1}{x} - \ln x}{e^x} = \dfrac{1 - x\ln x}{xe^x}$

240. $\dfrac{dy}{dx} = \dfrac{e^{2x}\cos x - 2e^{2x}\sin x}{e^{4x}} = \dfrac{\cos x - 2\sin x}{e^{2x}}$

241. $\dfrac{dy}{dx} = \dfrac{2x\ln x - x^2\cdot\frac{1}{x}}{(\ln x)^2} = \dfrac{x(2\ln x - 1)}{(\ln x)^2}$

242. $\dfrac{dy}{dx} = -e^x\sin(e^x)$

243. $\dfrac{dy}{dx} = \sec^2 x\, e^{\tan x}$

244. $\dfrac{dy}{dx} = \dfrac{e^x}{1 + e^x}$

245. $\dfrac{dy}{dx} = \dfrac{1 - \sin x}{x + \cos x}$

246. $\dfrac{dy}{dx} = \dfrac{e^x - e^{-x}}{e^x + e^{-x}}$

247. $\dfrac{dy}{dx} = \dfrac{2 + 2\cos 2x}{2x + \sin 2x}$

248. $\dfrac{dy}{dx} = 3\left(e^x+1\right)^2 \times e^x = 3e^x\left(e^x+1\right)^2$

249. $\dfrac{dy}{dx} = 2(\sin x + \cos x)(\cos x - \sin x)$

250. $y = (1+\ln x)^{\frac{1}{2}}$, so $\dfrac{dy}{dx} = \frac{1}{2}(1+\ln x)^{-\frac{1}{2}} \times \dfrac{1}{x} = \dfrac{1}{2x\sqrt{1+\ln x}}$.

251. $\dfrac{\cos x(1+\cos x)^2 + 2\sin^2 x(1+\cos x)}{(1+\cos x)^4} = \dfrac{\cos x + \cos^2 x + 2 - 2\cos^2 x}{(1+\cos x)^3} = \dfrac{(2-\cos x)(1+\cos x)}{(1+\cos x)^3} = \dfrac{2-\cos x}{(1+\cos x)^2}$

252. $\dfrac{dy}{dx} = \dfrac{\left(x^2+1\right)^2 - x\cdot 4x\left(x^2+1\right)}{\left(x^2+1\right)^4} = \dfrac{x^2+1-4x^2}{\left(x^2+1\right)^3} = \dfrac{1-3x^2}{\left(x^2+1\right)^3}$

253. $\dfrac{dy}{dx} = \dfrac{e^x\left(e^x+1\right)^3 - e^x\cdot 3e^x\left(e^x+1\right)^2}{\left(e^x+1\right)^6} = \dfrac{e^x\left(1-2e^x\right)}{\left(e^x+1\right)^4}$

254. $\dfrac{dy}{dx} = e^x(\sin x + \cos x) = 1$ at $x = 0$. The tangent is $y = x$.

255. Point $(1,\ \ln 2)$. $\dfrac{dy}{dx} = \dfrac{2x}{1+x^2} = 1$. $y = x - 1 + \ln 2$.

256. $\dfrac{dy}{dx} = e^{2x}(1+2x) = 1$ at $x = 0$, so the normal gradient is -1: $y = -x$.

257. In $e^x \sin x$ the last operation is multiplying two functions; in $e^{\sin x}$ one function is inside another. $\dfrac{d}{dx}e^x \sin x = e^x(\sin x + \cos x)$ and $\dfrac{d}{dx}e^{\sin x} = \cos x\, e^{\sin x}$.

258. $\dfrac{d}{dx}\ln(\sec x) = \dfrac{\tan x \sec x}{\sec x} = \tan x$

259. The student used the power rule on an exponential. For $e^{f(x)}$ the power stays the same and is multiplied by its derivative: $2xe^{x^2}$.

260. $\dfrac{dy}{dx} = \ln x + 1 - 1 = \ln x$, so the gradient at $x = e$ is 1.

261. a) $f'(x) = \dfrac{\frac{1}{x}\cdot x^2 - 2x\ln x}{x^4} = \dfrac{x(1-2\ln x)}{x^4} = \dfrac{1-2\ln x}{x^3}$ b) $\ln x = \frac{1}{2}$, so $x = \sqrt{e}$ and $f(\sqrt{e}) = \dfrac{\frac{1}{2}}{e} = \dfrac{1}{2e}$. The point is $\left(\sqrt{e},\ \dfrac{1}{2e}\right)$.

262. (a) $m = 2(-1) = -2$: $y = -2x - 1$. (b) AB has gradient 1; $2x = 1$, so $C\left(\frac{1}{2},\ \frac{1}{4}\right)$. (c) Tangent at B: $y = 4x - 4$. $-2x - 1 = 4x - 4$ gives $x = \frac{1}{2}$. *cf. 2010 HSC Q7(b)*

263. (a) $\dfrac{(1-x+x^2)(1+x)-1}{1+x} = \dfrac{1+x^3-1}{1+x} = \dfrac{x^3}{1+x}$ (b) $f'(x) - g'(x) = 1 - x + x^2 - \dfrac{1}{1+x} = \dfrac{x^3}{1+x} \geq 0$ for $x \geq 0$. (c) $g'(0) = 1$, $g(0) = 0$: $y = x$. *cf. 2009 HSC Q10(c)*

264. (a) $\sec^2 x - 1 = \tan^2 x$ (b) $\tan^2 x \geq 0$ for all x. (c) $f\left(\dfrac{\pi}{4}\right) = 1 - \dfrac{\pi}{4}$, $m = 1$: $y = x + 1 - \dfrac{\pi}{2}$.